MIL-M-28787/308A(EC)
23 July 1981
SUPERSEDING
MIL-M-28787/308(EC)
12 November 1980

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

PROGRAM CONTROL UNIT, KEY CODE MJC

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic program control unit module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN. ELEX 8111, Department of the Navy, Washington, D.C. 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.376 failures per million operating hours for class I or 0.927 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirement specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be MJC.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/308-1 for class I and M28787/308-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through LL shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

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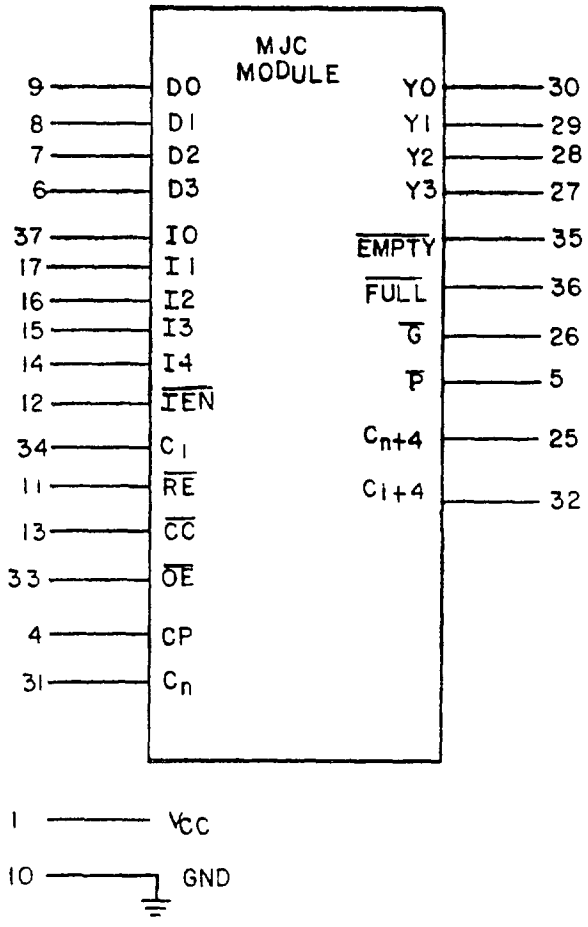


FIGURE 1. MJC module function diagram, program control unit.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	II	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	II	S	1	∞	1		1	∞	1	∞	1	∞	MΩ
B	Voltage (V_{CC}) stress	III	Q	← Must pass all remaining tests →										
C1-C16	Input breakdown current (I_{IB})	IV	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
D1-D16	Input clamp diode voltage	V	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
E1-E10	Output short circuit current (I_{OS})	VI	X	-30	-85	-30	-85	-30	-85	-30	-85	-30	-85	mA
F1-F4	Input current, high level	VII	X	0	20	0	20	0	20	0	20	0	20	µA
F5-F10, F12, F14, F15,	Input current, high level	VII	X	0	40	0	40	0	40	0	40	0	40	µA
F11	Input current, high level	VII	X	0	90	0	90	0	90	0	90	0	90	µA
F13	Input current, high level	VII	X	0	50	0	50	0	50	0	50	0	50	µA
F16	Input current, high level	VII	X	0	250	0	250	0	250	0	250	0	250	µA
G1-G4	Input current, low level	VIII	X	0.000	-0.398	0.000	-0.398	0.000	-0.398	0.000	-0.398	0.000	-0.398	mA
G5-G10, G12, G13, G15	Input current, low level	VIII	X	0.000	-0.775	0.000	-0.775	0.000	-0.775	0.000	-0.775	0.000	-0.775	mA
G11	Input current, low level	VIII	X	0.00	-2.55	0.00	-2.55	0.00	-2.55	0.00	-2.55	0.00	-2.55	mA
G14	Input current, low level	VIII	X	0.000	-0.725	0.000	-0.725	0.000	-0.725	0.000	-0.725	0.000	-0.725	mA
G16	Input current, low level	VIII	X	0.00	-3.59	0.00	-3.59	0.00	-3.59	0.00	-3.59	0.00	-3.59	mA
H1-H10	Output voltage, high level	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J10	Output voltage, low level	X	X	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	V

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
K1-K4	High impedance leakage current, high level	XI	X	-1	+50	-1	+50	-1	+50	-1	+50	-1	+50	µA
L1-L4	High impedance leakage current, low level	XII	X	-50	+1	-50	+1	-50	+1	-50	+1	-50	+1	µA
M	Power supply current	XIII	X	10	208	10	218	10	218	10	239	10	239	mA
N1-N6	Functional tests	XIV	X	← See table XIV and notes →										
P1-P10	Transition time, t_{TLH}	XVI	S	2	17	2	17	2	21	2	17	2	25	ns
Q1-Q10	Transition time, t_{THL}	XVII	S	1	18	1	18	1	21	1	18	1	24	ns
R1	Propagation delay time, t_{PLH}	XVIII	S	7	70	7	70	7	79	7	70	8	88	ns
R2,R3	Propagation delay time, t_{PLH}	XVIII	S	5	58	5	58	6	66	5	58	7	74	ns
R4	Propagation delay time, t_{PLH}	XVIII	S	6	65	6	65	7	73	6	65	8	82	ns
R5	Propagation delay time, t_{PLH}	XVIII	S	7	78	7	78	8	87	7	78	9	97	ns
R6	Propagation delay time, t_{PLH}	XVIII	S	6	69	6	69	7	78	6	69	8	87	ns
R7	Propagation delay time, t_{PLH}	XVIII	S	5	53	5	53	6	60	5	53	6	68	ns
R8,R11	Propagation delay time, t_{PLH}	XVIII	S	3	39	3	39	4	45	3	39	5	52	ns
R9	Propagation delay time, t_{PLH}	XVIII	S	4	46	4	46	5	53	4	46	6	60	ns
R10	Propagation delay time, t_{PLH}	XVIII	S	6	61	6	61	6	69	6	61	7	78	ns
R12	Propagation delay time, t_{PLH}	XVIII	S	2	29	2	29	3	33	2	29	3	37	ns
R13	Propagation delay time, t_{PLH}	XVIII	S	2	21	2	21	2	25	2	21	3	30	ns
R14	Propagation delay time, t_{PLH}	XVIII	S	3	36	3	36	4	41	3	36	4	46	ns
R15,R16	Propagation delay time, t_{PLH}	XVIII	S	2	17	2	17	2	20	2	17	2	23	ns
R17	Propagation delay time, t_{PLH}	XVIII	S	5	59	5	59	6	66	5	59	7	74	ns
R18,R19	Propagation delay time, t_{PLH}	XVIII	S	4	46	4	46	5	52	4	46	5	58	ns
R20	Propagation delay time, t_{PLH}	XVIII	S	5	53	5	53	5	59	5	53	6	66	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
R21	Propagation delay time, t_{PLH}	XVIII	S	6	66	6	66	7	75	6	66	8	84	ns
R22	Propagation delay time, t_{PLH}	XVIII	S	3	38	3	38	4	43	3	38	4	48	ns
R23,R24	Propagation delay time, t_{PLH}	XVIII	S	4	47	4	47	5	53	4	47	6	60	ns
R25	Propagation delay time, t_{PLH}	XVIII	S	4	43	4	43	4	49	4	43	5	55	ns
R26,R29	Propagation delay time, t_{PLH}	XVIII	S	2	28	2	28	3	33	2	28	3	38	ns
R27	Propagation delay time, t_{PLH}	XVIII	S	3	35	3	35	4	40	3	35	4	45	ns
R28	Propagation delay time, t_{PLH}	XVIII	S	5	50	5	50	5	57	5	50	6	65	ns
S1	Propagation delay time, t_{PHL}	XIX	S	7	70	7	70	7	79	7	70	8	88	ns
S2,S3	Propagation delay time, t_{PHL}	XIX	S	5	58	5	58	6	66	5	58	7	74	ns
S4	Propagation delay time, t_{PHL}	XIX	S	6	65	5	65	7	73	6	65	8	82	ns
S5	Propagation delay time, t_{PHL}	XIX	S	7	78	7	78	8	87	7	78	9	97	ns
S6	Propagation delay time, t_{PHL}	XIX	S	6	69	6	69	7	78	6	69	8	87	ns
S7	Propagation delay time, t_{PHL}	XIX	S	5	53	5	53	6	60	5	53	6	68	ns
S8,S11	Propagation delay time, t_{PHL}	XIX	S	3	39	3	39	4	45	3	39	5	52	ns
S9	Propagation delay time, t_{PHL}	XIX	S	4	46	4	46	5	53	4	46	6	60	ns
S10	Propagation delay time, t_{PHL}	XIX	S	6	61	6	61	6	69	6	61	7	78	ns
S12	Propagation delay time, t_{PHL}	XIX	S	2	29	2	29	3	33	2	29	3	37	ns
S13	Propagation delay time, t_{PHL}	XIX	S	2	21	2	21	2	25	2	21	3	30	ns
S14	Propagation delay time, t_{PHL}	XIX	S	3	36	3	36	4	41	3	36	4	46	ns
S15,S16	Propagation delay time, t_{PHL}	XIX	S	2	17	2	17	2	20	2	17	2	23	ns
S18,S19	Propagation delay time, t_{PHL}	XIX	S	4	46	4	46	5	52	4	46	5	58	ns
S20	Propagation delay time, t_{PHL}	XIX	S	5	53	5	53	5	59	5	53	6	66	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
S17	Propagation delay time, t_{PHL}	XIX	S	5	59	5	59	6	66	5	59	7	74	ns
S21	Propagation delay time, t_{PHL}	XIX	S	6	66	6	66	7	75	6	66	8	84	ns
S22	Propagation delay time, t_{PHL}	XIX	S	3	38	3	38	4	43	3	38	4	48	ns
S23, S24	Propagation delay time, t_{PHL}	XIX	S	4	47	4	47	5	53	4	47	6	60	ns
S25	Propagation delay time, t_{PHL}	XIX	S	4	43	4	43	4	49	4	43	5	55	ns
S26, S29	Propagation delay time, t_{PHL}	XIX	S	2	28	2	28	3	33	2	28	3	38	ns
S27	Propagation delay time, t_{PHL}	XIX	S	3	35	3	35	4	40	3	35	4	45	ns
S28	Propagation delay time, t_{PHL}	XIX	S	5	50	5	50	5	57	5	50	6	65	ns
T1-T4	Disable recovery time, t_{PLZ}	XX	S	5	55	5	55	6	60	5	55	6	65	ns
U1-U4	Enable recovery time, t_{PZL}	XXI	S	5	50	5	50	5	55	5	50	6	60	ns
V1-V4	Disable recovery time, t_{PHZ}	XXII	S	5	55	5	55	6	60	5	55	6	65	ns
W1-W4	Enable recovery time, t_{PZH}	XXIII	S	5	50	5	50	5	55	5	50	6	60	ns
X1-X9	Output crosstalk, low level	XXIV	Q	0	500	0	500	0	500	0	500	0	500	mV
Y1-Y8	Output crosstalk, high level	XXV	Q	0	500	0	500	0	500	0	500	0	500	mV
Z	Power supply transient amplitude	XXVI	Q	0	400	0	400	0	400	0	400	0	400	mV
AA	Power supply filter capacitance	XXVII	S	3.15	5.85	3.15	5.85	3.15	5.85	3.15	5.85	3.15	5.85	µF
BB1-BB4	Input capacitance	XXVIII	Q	6	55	6	55	6	55	6	55	6	55	pF
BB5-BB11, BB13-BB16	Input capacitance	XXVIII	Q	2	25	2	25	2	25	2	25	2	25	pF

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
BB12	Input capacitance	XXVIII	Q	4	40	4	40	4	40	4	40	4	40	pF
CC1-CC10	Output capacitance	XXIX	Q	2	30	2	30	2	30	2	30	2	30	pF
DD1-DD4	Enable recovery time, t_{PZL}	XXX	S	8	80	8	80	10	100	8	80	11	110	ns
EE1-EE4	Disable recovery time, t_{PL7}	XXXI	S	6	60	6	60	7	70	6	60	8	80	ns
FF1-FF4	Disable recovery time, t_{PH7}	XXXII	S	6	60	6	60	7	70	6	60	8	80	ns
GG1-GG4	Enable recovery time, t_{PZH}	XXXIII	S	8	80	8	80	10	100	8	80	11	110	ns
HH1-HH4	Enable recovery time, t_{PZL}	XXXIV	S	6	60	6	60	7	70	6	60	8	80	ns
JJ1-JJ4	Disable recovery time, t_{PLZ}	XXXV	S	5	50	5	50	6	60	5	50	7	70	ns
KK1-KK4	Enable recovery time, t_{PZH}	XXXVI	S	6	60	6	60	7	70	6	60	8	80	ns
LL1-LL4	Disable recovery time, t_{PHZ}	XXXVII	S	5	50	5	50	6	60	5	50	7	70	ns

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature test. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through LL in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through AA and DD through LL shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition AA, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through AA and DD through LL shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through LL shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Module description. The following electrical parameters reflect the maximum class temperature end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation:	1.199	1.315	W
Input voltage			
Low level	-0.5 min, 0.8 max	-0.5 min, 0.8 max	V
High level	2.0 min, V_{CC} max	2.0 min, V_{CC} max	V
Input current, low level (V_{IL} = 0.0 V):			
D0-D3	0.0 min, -0.398 max	0.0 min, -0.398 max	mA
I0-I4, $\overline{RE}$, $\overline{IEN}$, CP, $\overline{OE}$	0.0 min, -0.775 max	0.0 min, -0.775 max	mA
$\overline{CC}$	0.0 min, -0.725 max	0.0 min, -0.725 max	mA
C_1	0.0 min, -2.55 max	0.0 min, -2.55 max	mA
C_n	0.0 min, -3.59 max	0.0 min, -3.59 max	mA

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6.4 -Continued

	Class I	Class II	Unit
Input current, high level (V_{IH} = 2.7 V)			
D0-D3	0 min, 20 max	0 min, 20 max	µA
I0-I4, $\overline{RE}$, $\overline{IEN}$, CP, $\overline{OE}$	0 min, 40 max	0 min, 40 max	µA
$\overline{CC}$	0 min, 50 max	0 min, 50 max	µA
C_1	0 min, 90 max	0 min, 90 max	µA
C_n	0 min, 250 max	0 min, 250 max	µA
Clock parameters			
Low time	35 min	35 min	ns
High time	35 min	35 min	ns
Maximum frequency	14.2	14.2	MHz
Setup time (relative to clock low-to-high transition)			
I4-I0	124 min	124 min	ns
$\overline{CC}$	80 min	80 min	ns
$\overline{IEN}$	69 min	69 min	ns
C_n	52 min	52 min	ns
C_1	37 min	37 min	ns
D ($\overline{RE}$ = L, I4-I0 = 0-8 or 10-15)	30 min	30 min	ns
D (All other conditions)	72 min	72 min	ns
$\overline{RF}$	29 min	29 min	ns
Hold time (relative to clock low-to-high transition)			
I4-I0, $\overline{CC}$, $\overline{IEN}$, C_n	0 min	0 min	ns
C_1	5 min	5 min	ns
D ($\overline{RE}$ = L, I4-I0 = 0-8 or 10-15	2 min	2 min	ns
D (all other conditions)	2 min	2 min	ns
$\overline{RE}$	4 min	4 min	ns

6.5 Module function. The MIL-M-28787/308 module (MJC) is a four-bit wide program control unit (PCU) intended to perform machine level addressing functions, although the device can also be used as a microprogram sequencer. Four MIL-M-28787/308 modules may be interconnected to generate a 16-bit address (64k words). The MIL-M-28787/308 contains a program counter, a subroutine stack, an auxiliary register, and a full adder for computing relative addresses. The MIL-M-28787/308 performs five types of instructions

1. Unconditional fetch.
2. Conditional jump.
3. Conditional jump-to-subroutine.
4. Conditional return-from-subroutine.
5. Miscellaneous instructions.

There are four sources of data for the adder which generates the address outputs (Y0-Y3).

1. Program counter (PC).
2. Stack (S).
3. Auxiliary register (R).
4. Direct inputs (D).

Under control of the instruction inputs (I0-I4), the multiplexers at the adder inputs allow various combinations of these terms to be generated at the three-state Y address outputs. The instruction lines also control the updating of the program counter and the auxiliary register. A condition code input is provided for conditional instructions.

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6.6 Module use. The MIL-M-28787/301, 304, 305, and 308 (ERD, ZJU, ZJX, and MJC) modules are a series of designs, implementing the 2900 family as building blocks to emulate microprogrammed computers and/or controllers.

6.7 Definitions.

I0-I4 — The five instruction control lines to the Am2930, used to establish date paths and enable internal registers.

$\overline{IEN}$ — The instruction enable input, used to enable and disable internal registers. When $\overline{IEN}$ is low, all internal registers are under control of the instruction inputs. When $\overline{IEN}$ is high, all internal registers except R are inhibited from changing state. R is controlled by the $\overline{RE}$ input. The $\overline{IEN}$ input does not affect the combinatorial data paths and the outputs established by the instruction inputs.

$\overline{CC}$ — The condition code input determines whether or not a conditional instruction (instructions 16-31) is performed. If $\overline{CC}$ is low, the conditional instruction is executed. If $\overline{CC}$ is high, fetch PC (Instruction 1) is executed. The $\overline{CC}$ input may be either high or low for unconditional instructions (instructions 0-15).

$\overline{RE}$ — The register enable input for the auxiliary register (R). A low on $\overline{RE}$ causes the auxiliary register (R) to be loaded from the D inputs unless instruction 8 or 9 is being executed and $\overline{IEN}$ is low.

C_n — The carry-in to the full adder.

C_{n+4} — The carry-out of the full adder.

$\overline{P}$, $\overline{G}$ — The carry generate and propagate outputs of the full adder.

C_1 — The carry-in to the program counter incrementer

C_{1+4} — The carry-out of the program counter incrementer.

Y0-Y3 — The four address outputs of the Am2930. These are three-state output lines. When enabled they display the outputs of the full adder.

$\overline{OE}$ — Output enable. When $\overline{OE}$ is high, the Y outputs are off (high-impedance); when $\overline{OE}$ is low, the Y outputs are active (high or low).

D0-D3 — The four direct inputs which are used as inputs to the auxiliary register, the RAM, and the full adder, under instruction control.

$\overline{EMPTY}$ — The Empty output is low when the stack is empty.

$\overline{FULL}$ — The Full output is low when the LIFO stack is full - during and after the 17th push operation.

CP — The clock input to the Am2930. All internal registers (R, SP, PC) and the RAM are updated on the low-to-high transition of the clock input.

* 6.8 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 603) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and may be obtained upon request.

* 6.9 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

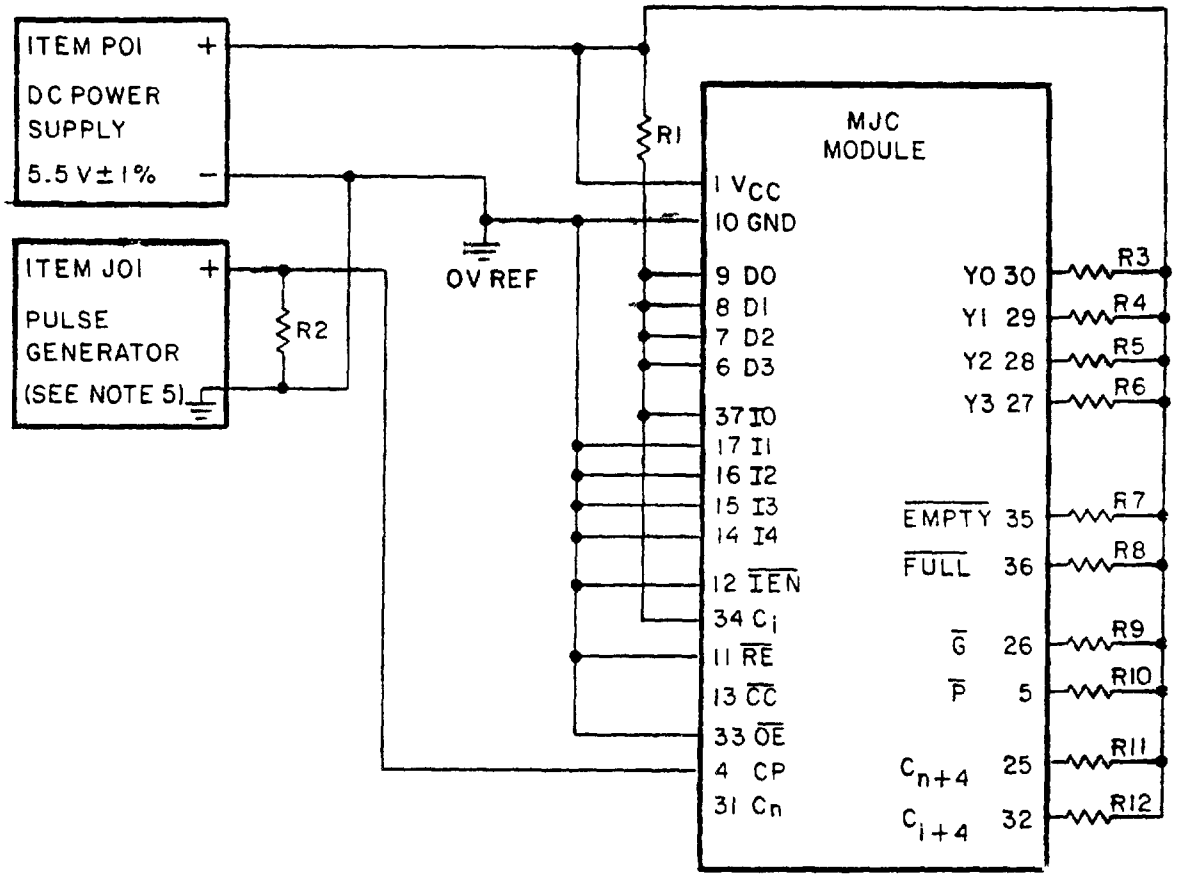
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Navy - EC

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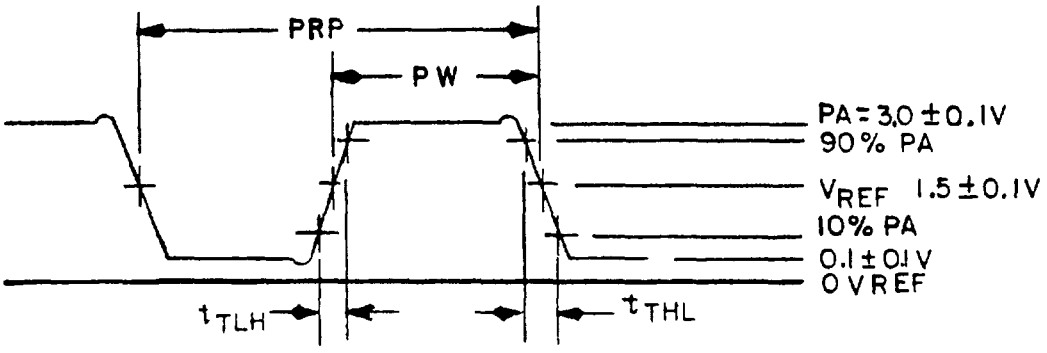
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NOTES.
1. R1 = 1 kΩ ±5%, 1/4 W.
2. R2 = 51Ω ±5%, 1/2 W.
3. R7, R8, R10 = 420Ω ±1%, 1/8 W.
4. R3-R6, R9, R11, R12 = 316Ω ±1%, 1/8 W.
5. Item J01 setup See figure 3.

FIGURE 2. Operating temperature and life test circuit.

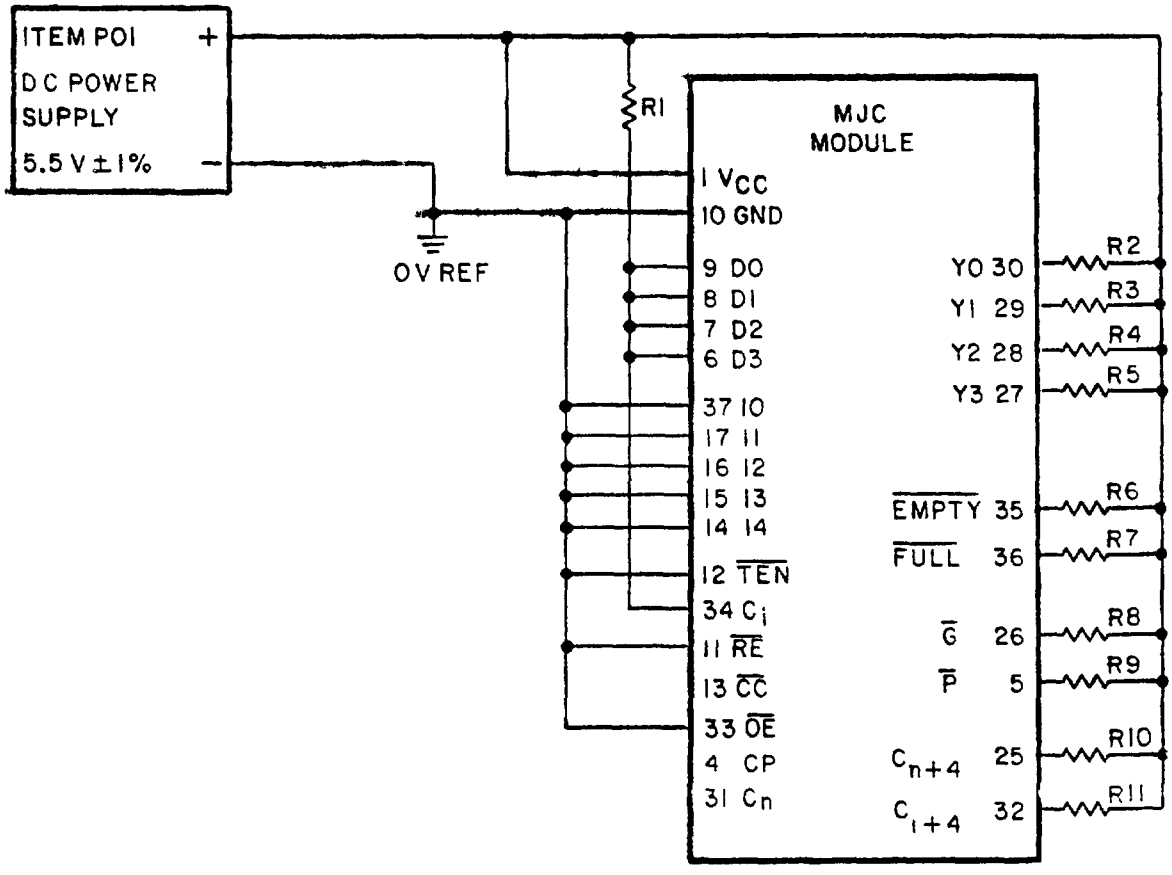
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NOTE: Input pulse parameters:
$t_{TLH} = t_{THL} = 10 \pm 2$ ns.
PW = 500 ±50 ns.
PRP = 1,000 ±100 ns.

*FIGURE 3. Input pulse waveform for figures 2, 10, and 12.

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NOTES
1. R1 = 1 kΩ ±5%, 1/4 W.
2. R2-R5, R8, R10, R11 = 316Ω ±1%, 1/8 W.
3. R6, R7, R9 = 420Ω ±1%, 1/8 W.

*FIGURE 4. Component temperature test circuit.

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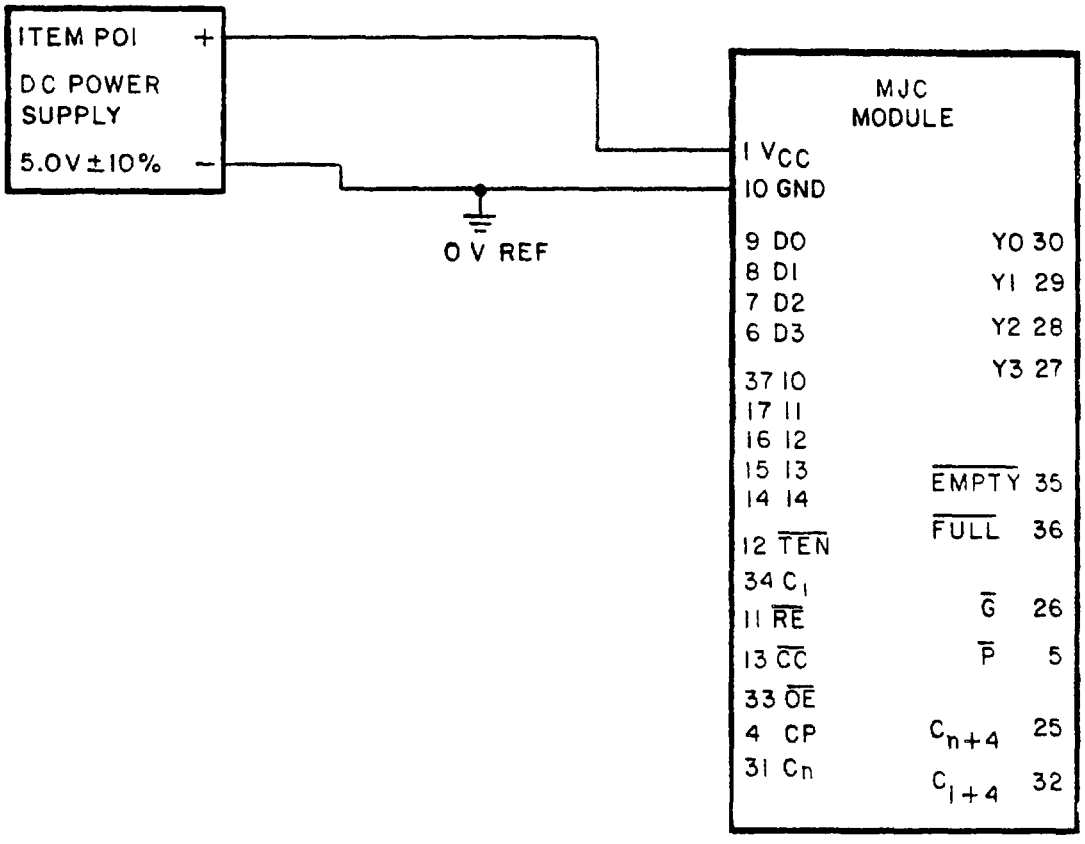


*FIGURE 5. Vibration test circuit.

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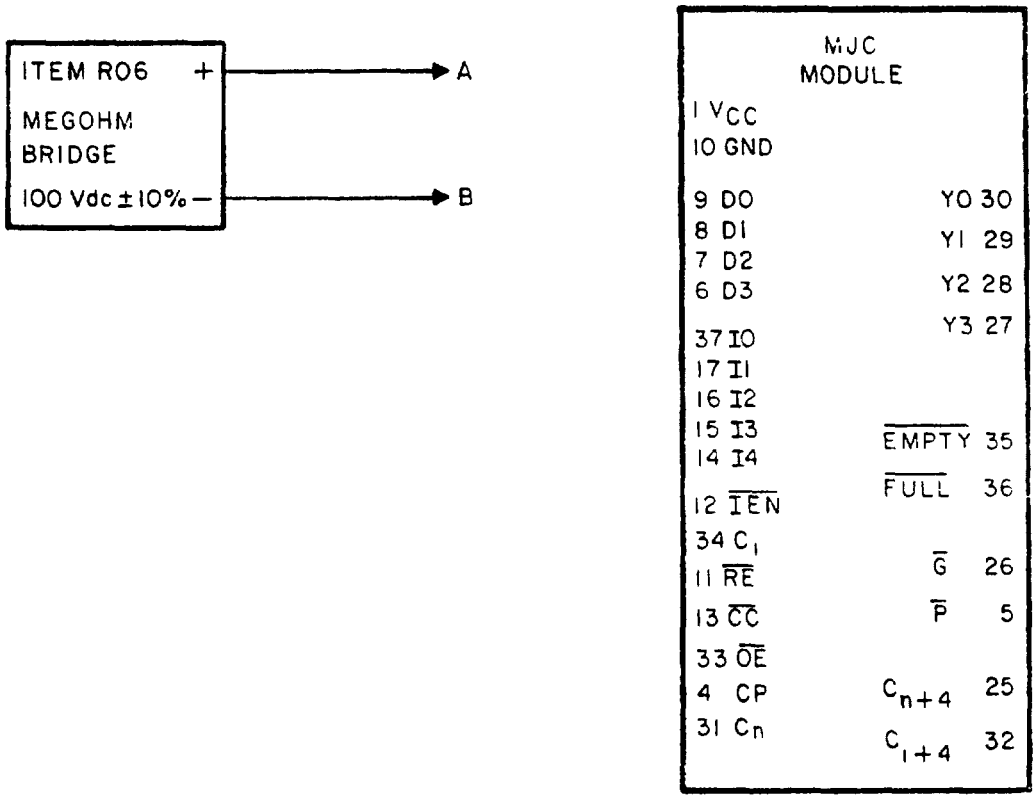
TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	all pins	Frame	10	∞	MΩ	6

TABLE III. Voltage (V_{CC}) stress test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
			Min	Max		
B	Voltage (V_{CC}) stress	←Must pass remainder of tests ⟶				7

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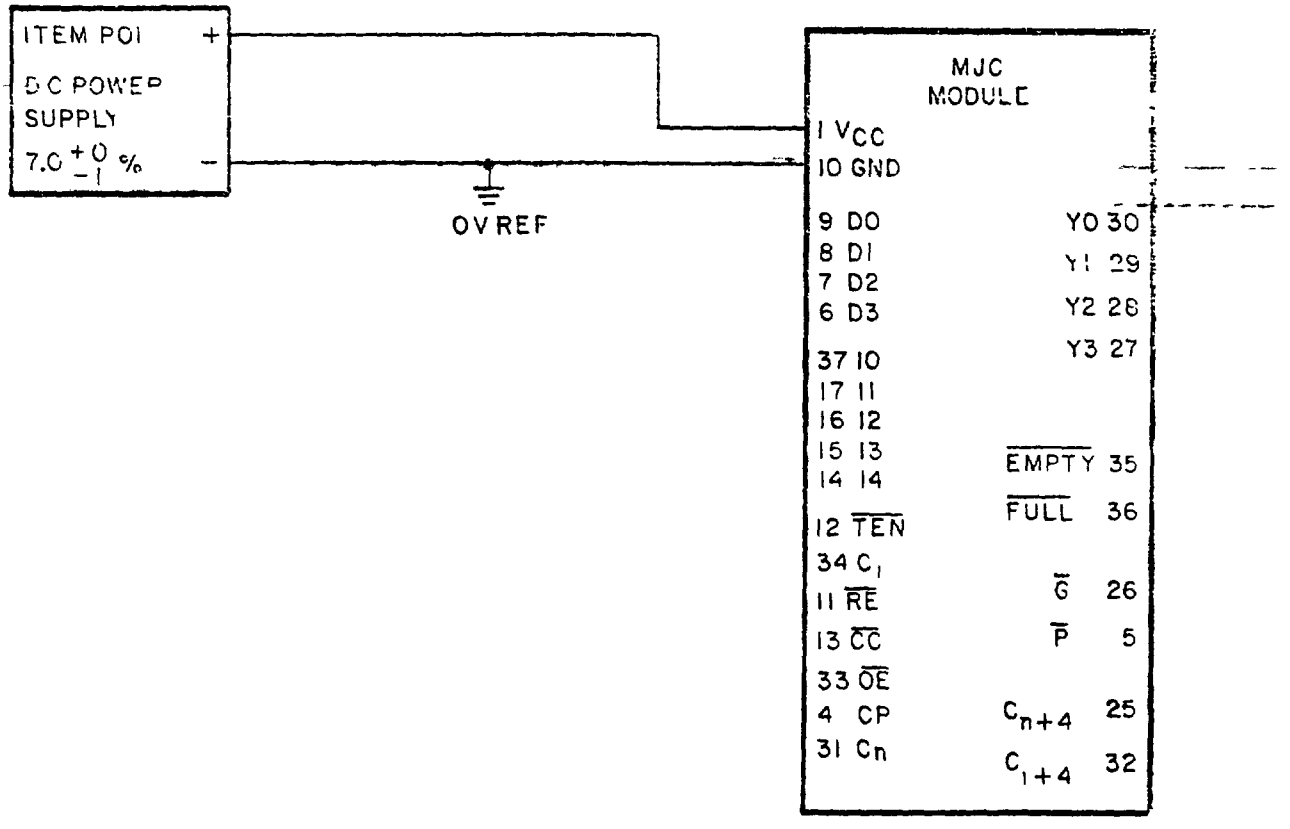
WARNING:

When using the megohm bridge, high voltage may be present at any of the output binding posts Always set the function switch of the megohm bridge to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 6. Isolation test circuit for table II.

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NOTE Maintain test conditions for a minimum of 10 seconds.

*FIGURE 7. Voltage (V_{CC}) stress test circuit for table III.

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TABLE IV. Input breakdown current (I_{IB}) test.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C1	D0	Input breakdown current (I_{IB})	9	6,7,8,11,12,13,14,15,16,17,4,31,33,34,37	0.0	1.0	mA	8
C2	D1		8	6,7,9,11,12,13,14,15,16,17,4,31,33,34,37				
C3	D2		7	6,8,9,11,12,13,14,15,16,17,4,31,33,34,37				
C4	D3		6	7,8,9,11,12,13,14,15,16,17,4,31,33,34,37				
C5	I0		37	6,7,8,9,11,12,13,14,15,16,17,4,31,33,34				
C6	I1		17	6,7,8,9,11,12,13,14,15,16,4,31,33,34,37				
C7	I2		16	6,7,8,9,11,12,13,14,15,17,4,31,33,34,37				
C8	I3		15	4,6,7,8,9,11,12,13,14,16,17,31,33,34,37				
C9	I4		14	4,6,7,8,9,11,12,13,15,16,17,31,33,34,37				
C10	$\overline{IEN}$		12	4,6,7,8,9,11,13,14,15,16,17,31,33,34,37				
C11	C_i		34	4,6,7,8,9,11,12,13,14,15,16,17,31,33,37				
C12	$\overline{RE}$		11	4,6,7,8,9,12,13,14,15,16,17,31,33,34,37				

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TABLE IV. Input breakdown current (I_{IB}) test - Continued.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C13	$\overline{CC}$	Input breakdown current (I_{IB})	13	4,6,7,8,9,11,12,14,15, 16,17,31,33,34,37	0.0	1.0	mA	8
C14	$\overline{OE}$	↓	33	4,6,7,8,9,11,12,13,14, 15,16,17,31,34,37	↓	↓	↓	↓
C15	CP	↓	4	6,7,8,9,11,12,13,14,15, 16,17,31,33,34,37	↓	↓	↓	↓
C16	C_n	↓	31	4,6,7,8,9,11,12,13,14, 15,16,17,33,34,37	↓	↓	↓	↓

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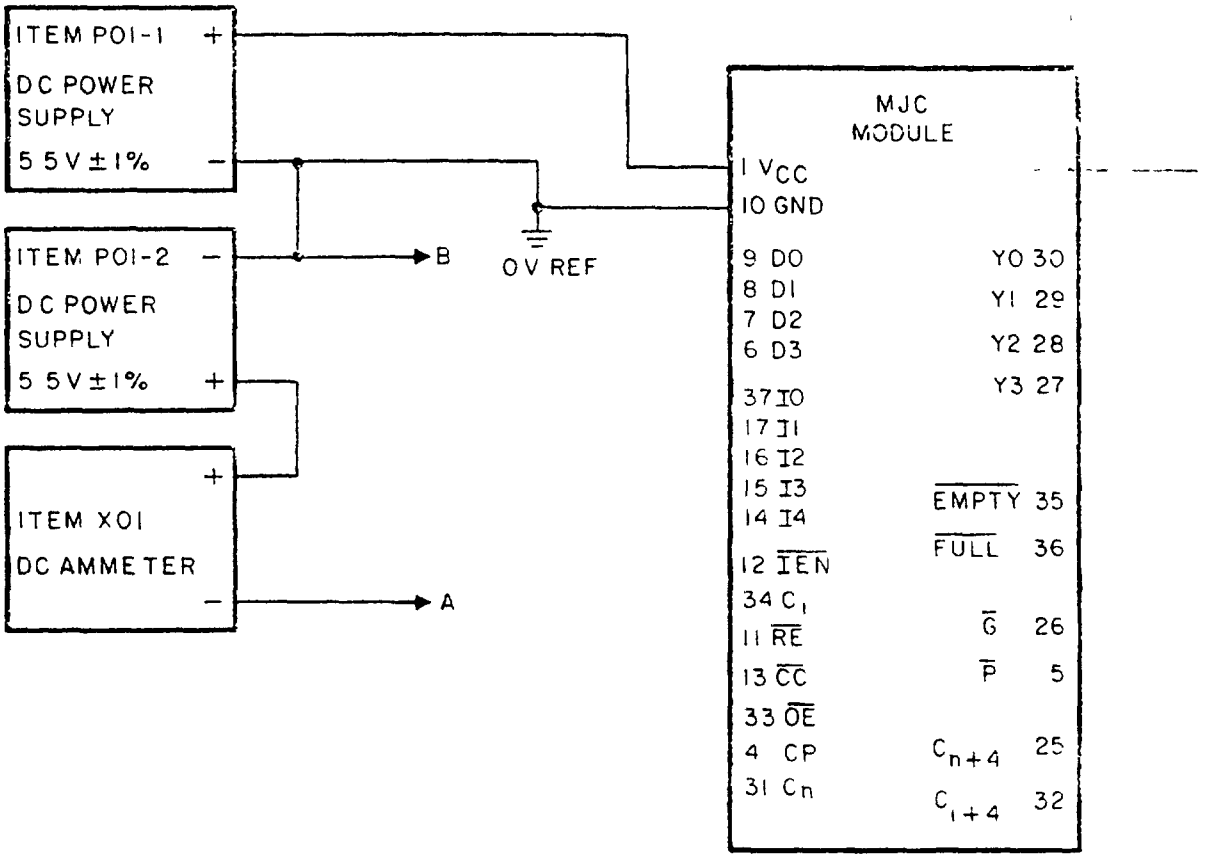


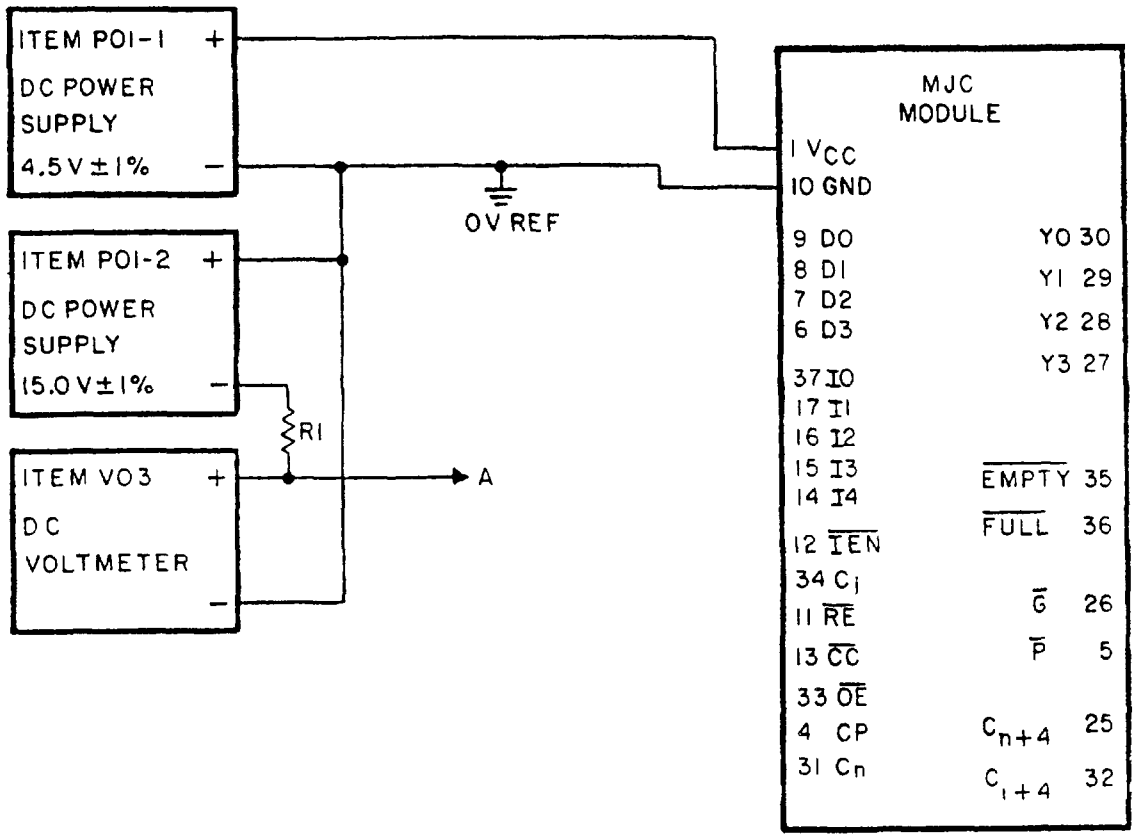
FIGURE 8. Input breakdown current test circuit for table IV

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* TABLE V. Input clamp diode voltage test.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
D1	D0	Input clamp diode voltage (V_{IC})	9	-0.4	-1.5	V	9
D2	D1	↓	8	↓	↓	↓	↓
D3	D2		7				
D4	D3		6				
D5	I0		37				
D6	I1		17				
D7	I2		16				
D8	I3		15				
D9	I4		14				
D10	$\overline{IEN}$		12				
D11	C_i		34				
D12	$\overline{RE}$		11				
D13	$\overline{CC}$		13				
D14	$\overline{OE}$		33				
D15	CP		4				
D16	C_n		31				

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NOTE: R1 = 750Ω ±1%, 1/4 W.

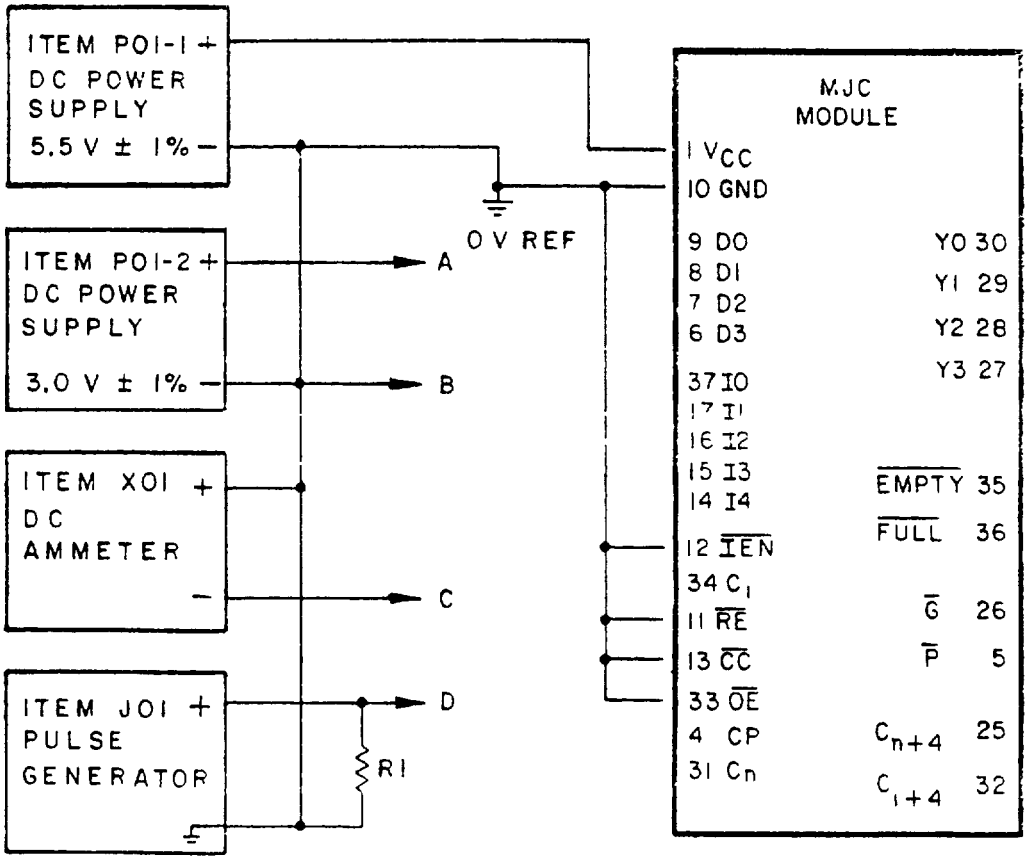
FIGURE 9. Input clamp diode voltage test circuit for table V

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TABLE VI. Output short circuit current (I_{OS}) test.

Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
---	Setup		6,7,8,9	14,15,16,17, 37,34,31	Open	4	---	---	---	10
E1	Y0	Output short circuit current (I_{OS})	6,7,8,9,14, 34,37	15,16,17,31	30	Open	-30	-85	mA	
E2	Y1				29					
E3	Y2				28					
E4	Y3				27					
E5	$\overline{FULL}$				36					
E6	C_{n+4}				25					
E7	C_{i+4}				32					
---	Setup		6,7,8,9	14,15,16,17, 37,31,34	Open	4	---	---	---	
E8	$\overline{EMPTY}$	Output short circuit current (I_{OS})	6,7,8,9,15, 17,37	14,16,31,34	35	4	-30	-85	mA	
---	Setup		6,7,8,9	14,15,16,17, 37,31,34	Open	4	---	---	---	
E9	$\overline{G}$	Output short circuit current (I_{OS})	14,16,7,8,9	6,15,17,31, 34,37	26	Open	-30	-85	mA	
E10	$\overline{P}$	Output short circuit current (I_{OS})	7,8,9,16,14	6,15,17,31, 34,37	5	Open	-30	-85	mA	

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NOTES
1. R1 = 51Ω ±5%, 1/2 W.
2. Item J01 is used in the single pulse mode When point D is referenced, apply a single pulse per figure 3.
3. Duration of test should not exceed 1 second

FIGURE 10. Output short circuit current (I_{OS}) test circuit for table VI.

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* TABLE VII Input current (high-level) test.

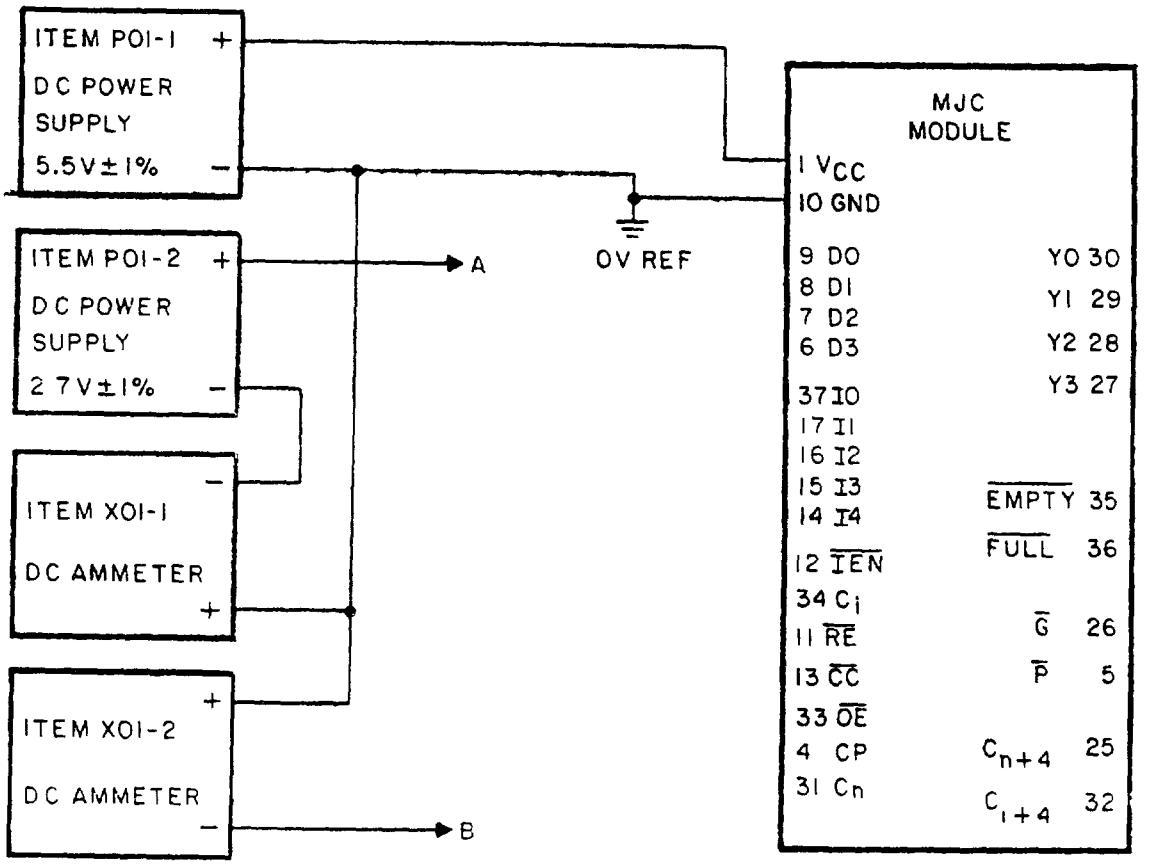
Test condition	Parameter	Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
F1	DO Input current, high level	9	Open	0	20	µA	11
F2	D1	8					
F3	D2	7					
F4	D3	6					
F5	IO	37			40		
F6	I1	17					
F7	I2	16					
F8	I3	15					
F9	I4	14					
F10	$\overline{IFN}$	12					
F11	C_i	34			90		
F12	$\overline{RE}$	11			40		
F13	$\overline{CC}$	13			50		
F14	$\overline{OE}$	33			40		
F15	CP	4			40		
F16	C_n	31			250		

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* TABLE VIII. Input current (low-level) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
G1	D0 Input current, low level	Open	9	0.000	-0.398	mA	11
G2	D1		8				
G3	D2		7				
G4	D3		6				
G5	I0		37		-0.775		
G6	I1		17				
G7	I2		16				
G8	I3		15				
G9	I4		14				
G10	$\overline{IEN}$		12				
G11	C_i		34	0.00	-2.55		
G12	$\overline{RE}$		11	0.000	-0.775		
G13	$\overline{CC}$		13		-0.775		
G14	$\overline{OE}$		33		-0.725		
G15	CP		4		-0.775		
G16	C_n		31	0.00	-3.59		

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NOTE Since items XO1-1 and XO1-2 are not used simultaneously, only one item XO1 is required.

FIGURE 11. Input current (high-level and low-level) test circuit for tables VII and VIII.

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TABLE IX. Output voltage (high-level) test.

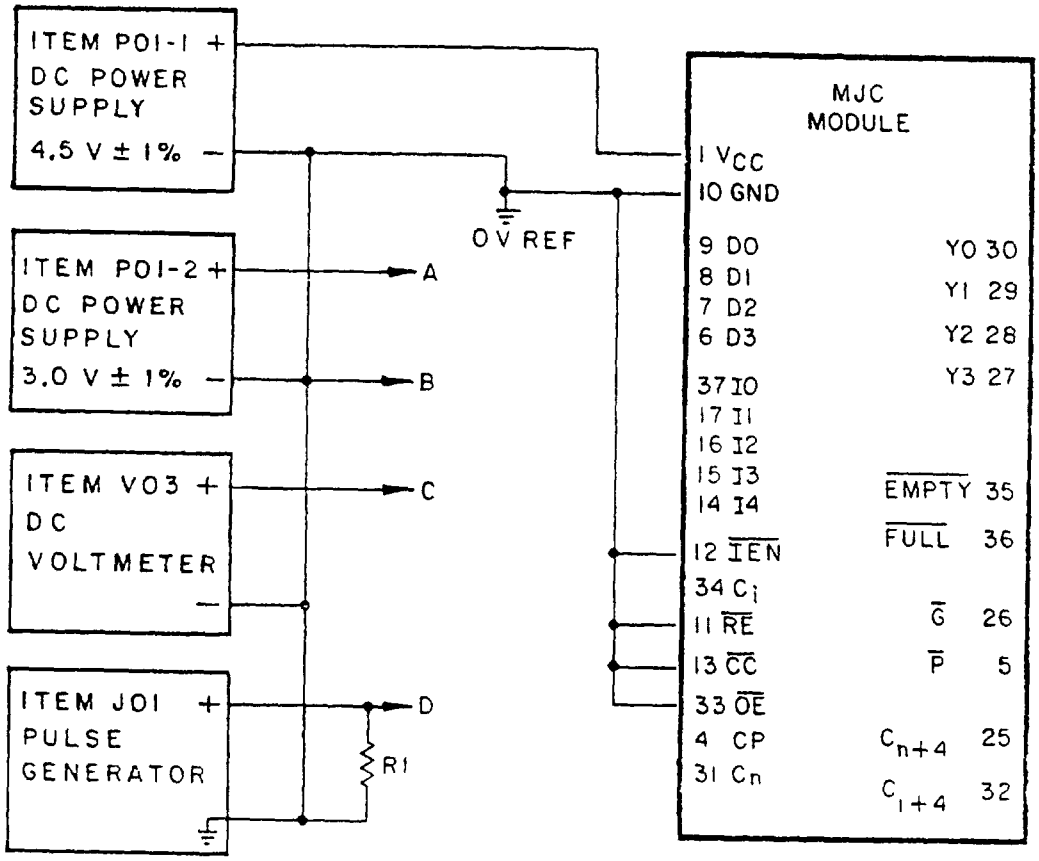
Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
---	Setup		6,7,8,9	14,15,16,17,37 31,34	Open	4	---	---	---	12
H1	YO	Output voltage, high level	6,7,8,9,14, 34,37	15,16,17,31	30	Open	2.4	4.5	V	
H2	Y1				29					
H3	Y2				28					
H4	Y3				27					
H5	$\overline{FULI}$				36					
H6	C_{n+4}				25					
H7	C_{1+4}				32					
---	Setup		6,7,8,9	14,15,16,17, 37,31,34	Open	4	---	---	---	
H8	$\overline{EMPTY}$	Output voltage, high level	15,37,17,6, 7,8,9	14,16,31,34	35	4	2.4	4.5	V	
---	Setup		6,7,8,9	14,15,16,17, 37,31,34	Open	4	---	---	---	
H9	$\overline{G}$	Output voltage, high level	14,16,7,8,9	6,15,17,31, 34,37	26	Open	2.4	4.5	V	
H10	$\overline{P}$	Output voltage, high level	7,8,9,14,16	6,15,17,31, 34,37	5	Open	2.4	4.5	V	7

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* TABLE X. Output voltage (low-level) test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
---	Setup	Open	14,15,16,17,31, 34,37	Open	4	---	---	---	17
J1	Y0 Output voltage, low level			30	Open	0.0	0.5	V	
J2	Y1			29					
J3	Y2			28					
J4	Y3			27					
J5	$\overline{\text{EMPTY}}$			35					
J6	$\overline{\text{G}}$			26					
J7	$\overline{\text{P}}$			5					
J8	C_{i+4}			32					
---	Setup			Open	4	---	---	---	
J9	$\overline{\text{FULL}}$ Output voltage, low level	15,17,37	14,16,31,34	36	Open	0.0	0.5	V	
---	Setup	Open	14,15,16,17,31, 34,37	Open	4	---	---	---	
J10	C_{n+4} Output voltage, low level	15,17,37	14,16,31,34	25	Open	0.0	0.5	V	

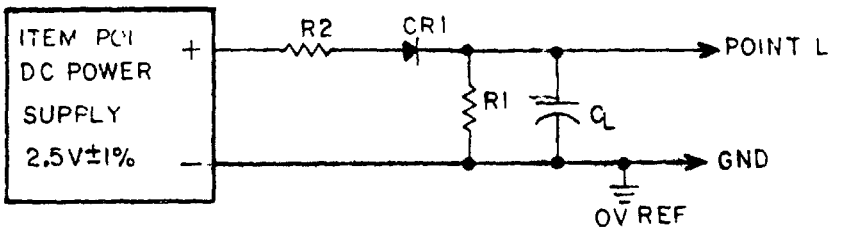
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. Load all outputs per figure 13.
3. Item J01 is used in the single pulse mode. When point D is referenced, apply a single pulse per figure 3.
4. For test condition J9, apply 17 single pulses after point D is referenced

FIGURE 12. Output voltage (high-level and low-level) test circuit for tables IX and X.

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NOTES
1. Load for outputs Y0, Y1, Y2, Y3, $\overline{G}$, C_{n+4} and C_{i+4}:
   CR1 = 1N4150 or equivalent
   C_L = 50 pF ±5% (including probe and parasitic capacitance) for test conditions P, Q, R, S, T, U, V, W, X, Y, DD, EE, FF, GG, HH, JJ, KK, and LL.
   R1 = 1.5 kΩ ±1%, 1/8 W.
   R2 = Select value for a current flow of 16 mA ±5% out of point L with point L held at 0.5 V.
2. Load for outputs $\overline{P}$, $\overline{FULL}$, and $\overline{EMPTY}$:
   CR1 = 1N4150 or equivalent
   C_L = 50 pF ±5% (including probe and parasitic capacitance) for test conditions P, Q, R, S, T, U, V, W, X, Y, DD, EE, FF, GG, HH, JJ, KK, and LL
   R1 = 2.0 kΩ ±1%, 1/8 W
   R2 = Select value for a current flow of 12 mA ±5% out of point L with point L held at 0.5 V
3. C_L is not required for dc electrical tests

FIGURE 13. MOC logic load circuit.

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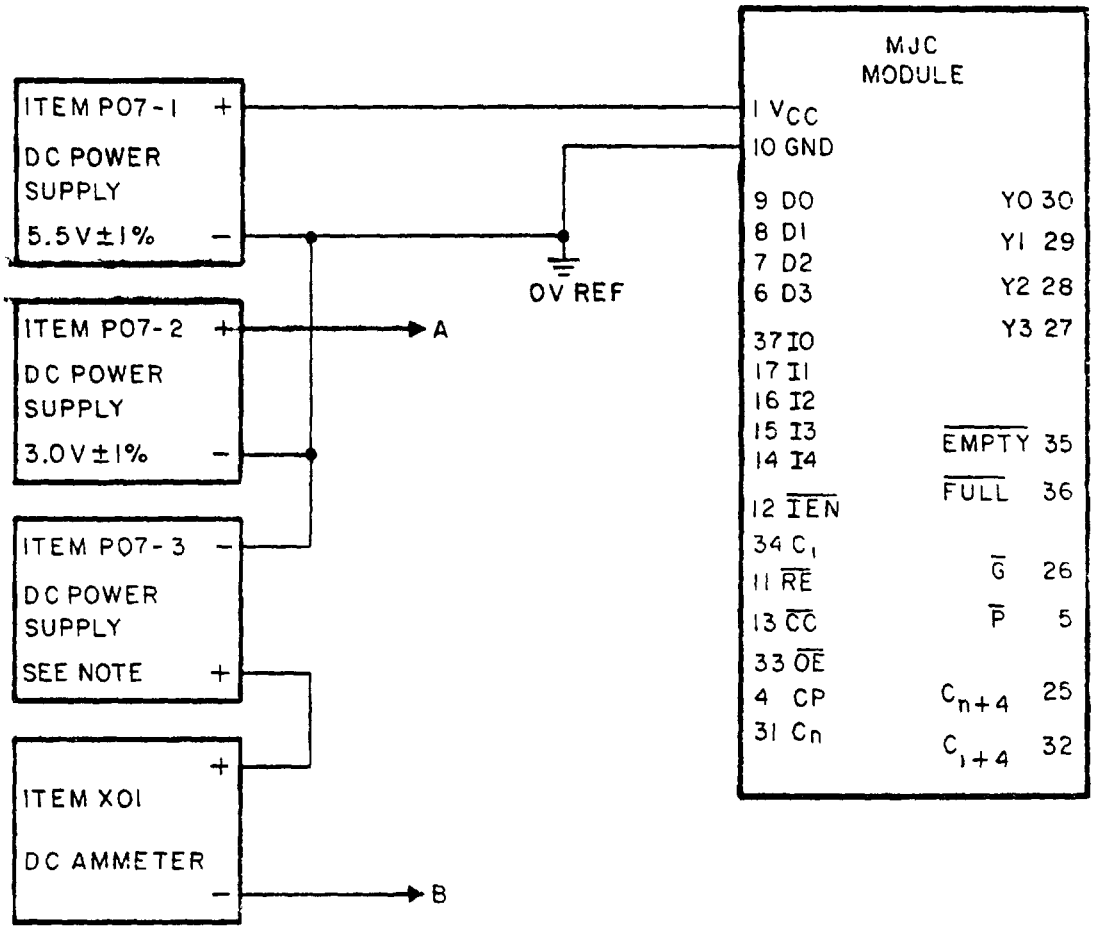
TABLE XI. High-impedance leakage current (high-level) test.

Test condition		Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
K1	Y0	High-impedance leakage current, high level	33	30	Open	-1	+50	µA	14
K2	Y1	↓	↓	29	↓	↓	↓	↓	↓
K3	Y2	↓	↓	28	↓	↓	↓	↓	↓
K4	Y3	↓	↓	27	↓	↓	↓	↓	↓

TABLE XII. High-impedance leakage current (low-level) test.

Test condition		Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
L1	Y0	High-impedance leakage current, low level	33	30	-50	+1	µA	14
L2	Y1	↓	↓	29	↓	↓	↓	↓
L3	Y2	↓	↓	28	↓	↓	↓	↓
L4	Y3	↓	↓	27	↓	↓	↓	↓

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NOTE: Set item P07-3 to 2.4 V ±1% for test condition K and 0.5 V ±1% for test condition L.

FIGURE 14. High impedance leakage current (high-level and low-level) test circuit for tables XI and XII.

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TABLE XIII. Power supply current test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M	Power supply current	6,7,8,9	13,14,15,16,17,37, 12,33	10	208	mA	15

* TABLE XIV. Dynamic functional test.

Test condition	Parameter	Test description	Initial limits 25° ±5°C ambient	Test figure
N1	Functional	Minimum clock pulse, low time (V_{CC} = 4.5 V)	1/, 11/	16
N2		Minimum clock pulse, low time (V_{CC} = 5.5 V)	2/, 11/	16
N3		Minimum clock pulse, high time (V_{CC} = 4.5 V)	3/, 11/	17
N4		Minimum clock pulse, high time (V_{CC} = 5.5 V)	4/, 11/	17
N5		Setup and hold time (V_{CC} = 4.5 V)	5/-11/	18,19
N6		Setup and hold time (V_{CC} = 5.5 V)	5/-11/	18,19

1/ Test condition N1 checks minimum clock pulse, low time with V_{CC} = 4.5 V. This test also checks hold time on $\overline{RE}$ only and all setup times except data to auxillary register. FPAT1 through FPAT12 checks low-to-high transition setup times and high-to-low transition hold times. FPAT13 through FPAT24 checks high-to-low transition setup times and low-to-high transition hold times.

2/ Test condition N2 is the same as test condition N1, except V_{CC} = 5.5 V.

3/ Test condition N3 is the same as test condition N1, except minimum clock pulse high time is checked.

4/ Test condition N4 is the same as test condition N3, except V_{CC} = 5.5 V.

5/ Test condition N5, part 1 checks hold time on data, instruction, $\overline{CC}$, $\overline{IEN}$, and C_n to the auxillary register with V_{CC} = 4.5 V. FPAT25 checks high-to-low transitions and FPAT26 checks low-to-high transitions. Test condition N6, part 1 is the same as test condition N5, part 1, except V_{CC} = 5.5 V.

6/ Test condition N5, part 2 checks hold time on data, instruction, $\overline{CC}$, and $\overline{IEN}$ inputs to the stack with V_{CC} = 4.5 V. See footnote 9 for pattern. This pattern can check both low-to-high and high-to-low transition in one pass. Test condition N6, part 2 is the same as test condition N5, part 2, except V_{CC} = 5.5 V.

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TABLE XIV - Continued

7/ Test condition N5, part 3 checks hold time on data, instruction, $\overline{CC}$, $\overline{IEN}$, and C_1 inputs to the program counter with V_{CC} = 4.5 V. FPAT27 checks high-to-low transitions and low-to-high transitions. Test condition N6, part 3 is the same as test condition N5, part 3, except V_{CC} = 5.5 V.

8/ Test condition N5, part 4 checks setup time on data inputs to the auxillary register with V_{CC} = 4.5 V. See footnote 10 for pattern. Test condition N6, part 4 is the same as test condition N5, part 4, except V_{CC} = 5.5 V.

9/ Pattern for test condition N5 (and N6), part 2. Instruction, data, and Y3-Y0 are listed in hexadecimal. X = don't care.

I4-I0	$\overline{CC}$	$\overline{IEN}$	C_1	$\overline{RE}$	$\overline{OE}$	D3-D0	Y3-Y0
LC	L	L	L	H	L	L	X
LD	L	L	L	H	L	L	L
LC	L	L	L	H	L	F	X
LD	L	L	L	H	L	F	F

10/ Pattern for test condition N5 (and N6), part 4. I4-I0, D3-D0, and Y3-Y0 are listed in hexadecimal.

I4-I0	$\overline{CC}$	$\overline{IEN}$	C_1	$\overline{RE}$	$\overline{OE}$	D3-D0	Y3-Y0
L2	L	L	L	L	L	F	F
L2	L	L	L	L	L	L	L
L2	L	L	L	L	L	F	F
L2	L	L	L	L	L	L	L

11/ Test conditions N1 through N6 apply the functional test vectors (see 6.8). Each test condition tests mainly for the test parameters as described in the test description, but also for the following operating conditions and parameters

a. Loaded output voltage levels (loaded per figure 13)
   V_{OH} = 2.4 V minimum and V_{OL} = 0.5 V maximum.
b. Worst case input threshold voltage levels.
   V_{IL} = 0.5 V maximum and V_{IH} = 2.4 V minimum.
c. Most positive and negative power supply tolerances, most negative = 4.5 V and most positive = 5.5 V.
d. Worst case input setup and hold times as specified in table XV.

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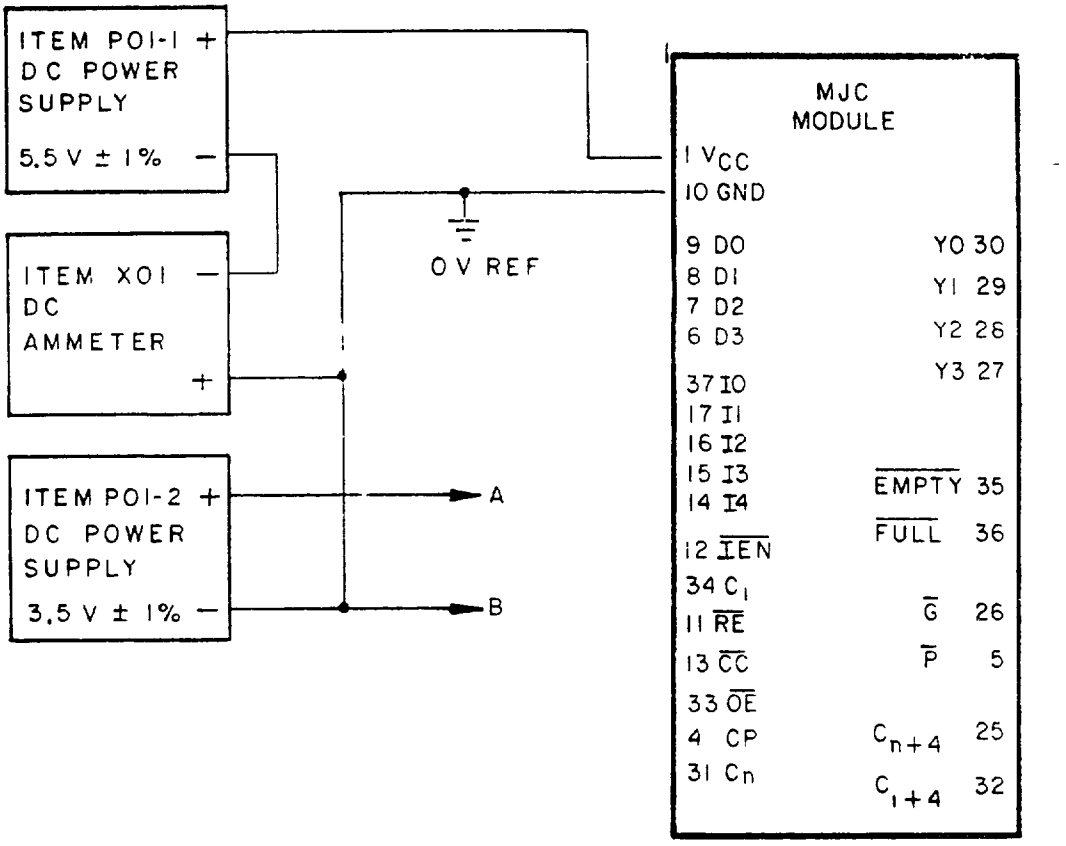
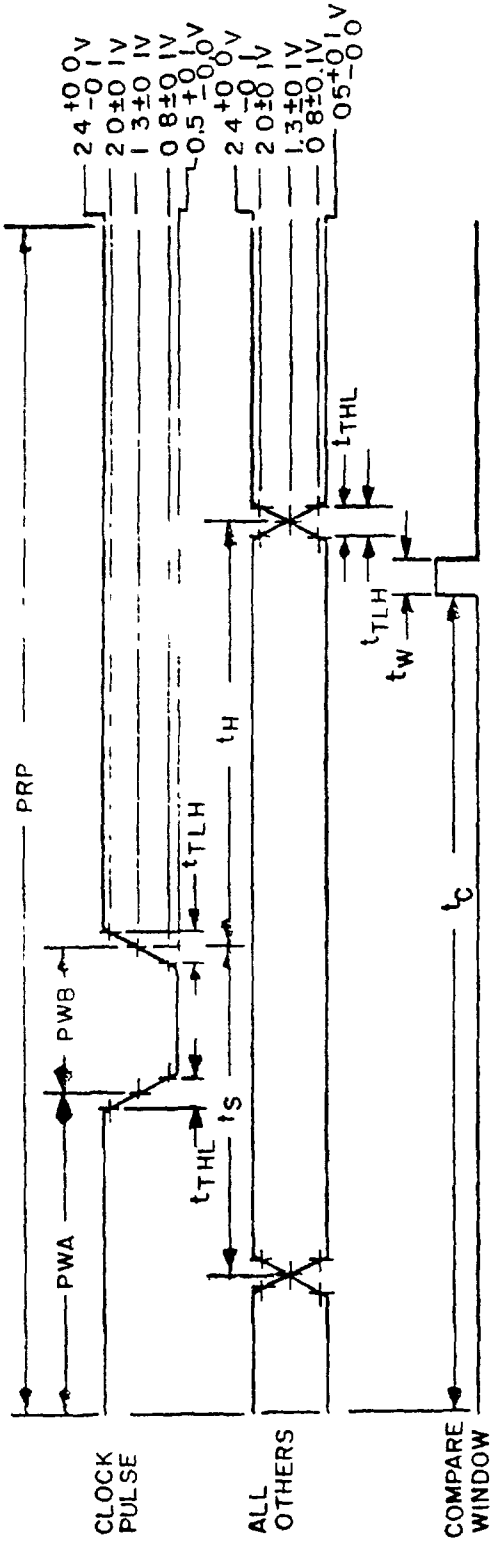


FIGURE 15. Power supply current test circuit for table XIII.

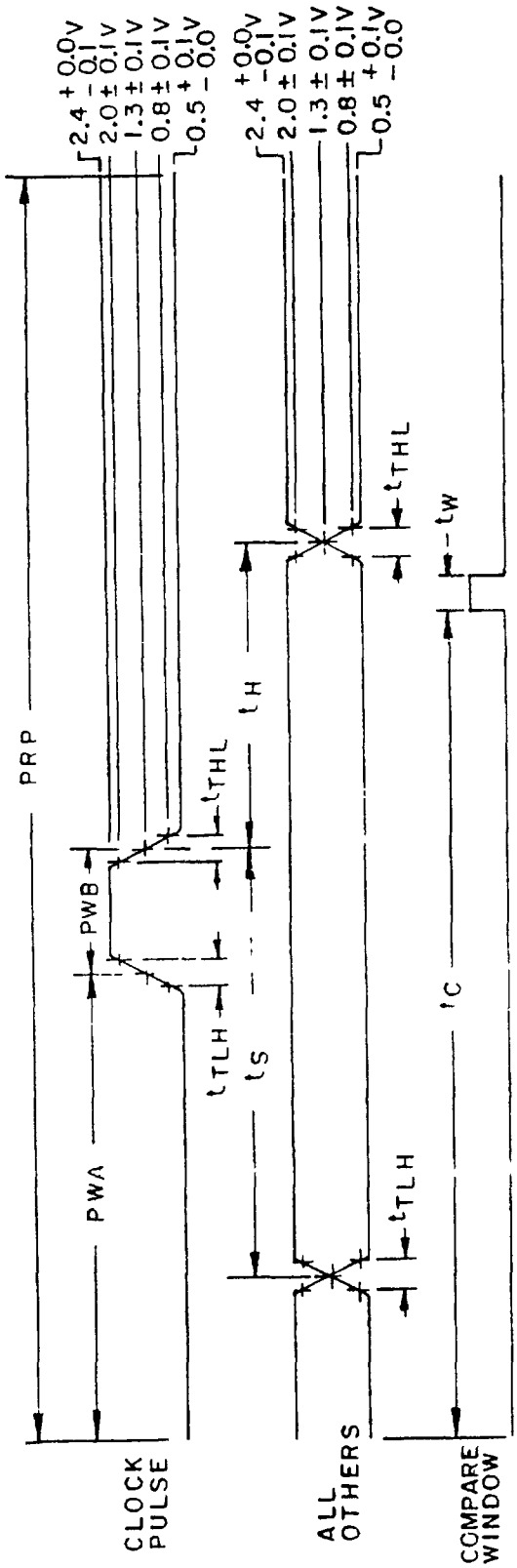
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NOTE: Pulse parameters:
PWA = 95 ±2 ns.
PWB = 36 ±2 ns.
PRP = 376 ±2 ns.
t_C = 350 2 ns (FULL); 230 ±2 ns (All others).
t_W = 10 ±2 ns.
t_{TLH} = t_{THL} = 5 ±5 ns
t_S = 125 ±1 ns (I0-I4).
81 ±1 ns ($\overline{CC}$).
70 ±1 ns ($\overline{IEN}$).
53 ±1 ns (C_n).
38 ±1 ns (C_4).
74 ±1 ns (D0-D3)
30 ±1 ns ($\overline{RE}$).
t_H = 5 +1 ns ($\overline{RE}$).
120 ±5 ns (All others).

*FIGURE 18 Dynamic functional test (minimum clock pulse low) waveforms for table XIV.

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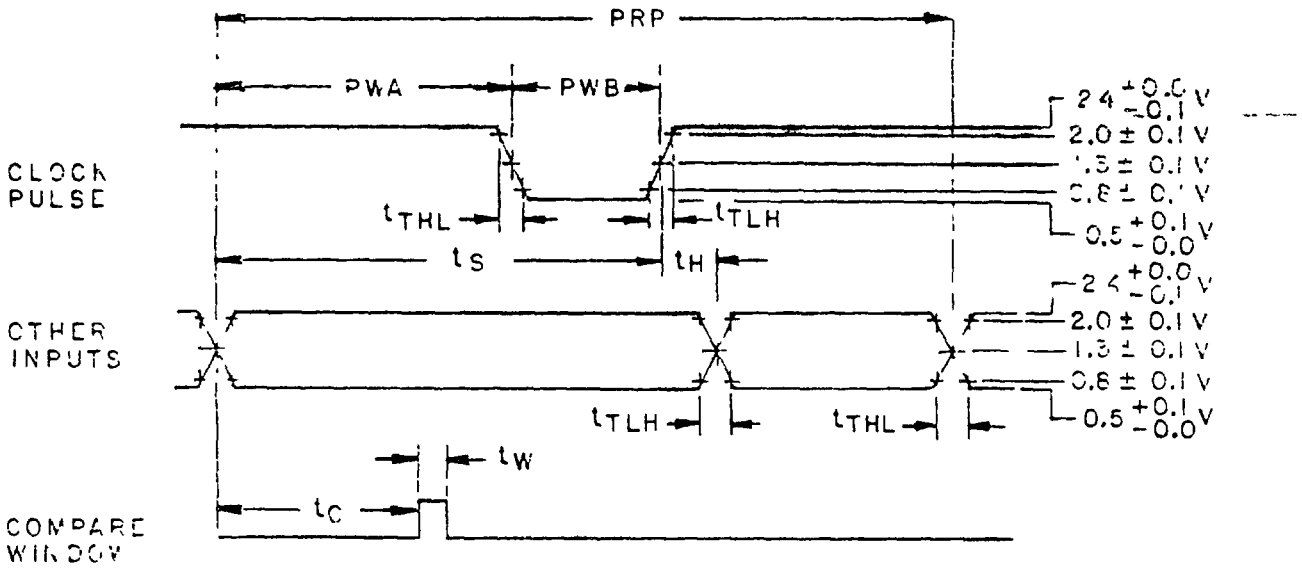


NOTE Pulse parameters
PRP = 376 ±2 ns
PWA = 95 ±2 ns.
PWB = 36 ±2 ns. ($\overline{FULL}$)
t_C = 350 ±2 ns (All others).
t_C = 230 ±2 ns (All others).
t_W = 10 ±2 ns.
$t_{TLH} = t_{THL}$ = 5 ±5 ns
t_H = 5 ±1 ns ($\overline{RE}$).
t_H = 120 ±2 ns (All others).

t_S = 125 ±1 ns (I0-I4)
= 81 ±1 ns ($\overline{CC}$).
= 70 ±1 ns ($\overline{TEN}$).
= 53 ±1 ns (C_n).
= 38 ±1 ns (C_{n+1}).
= 74 ±1 ns ($\overline{D0}$-D3)
= 30 ±1 ns ($\overline{RE}$).

FIGURE 17 Dynamic functional test (minimum clock pulse high) waveforms for table XIV.

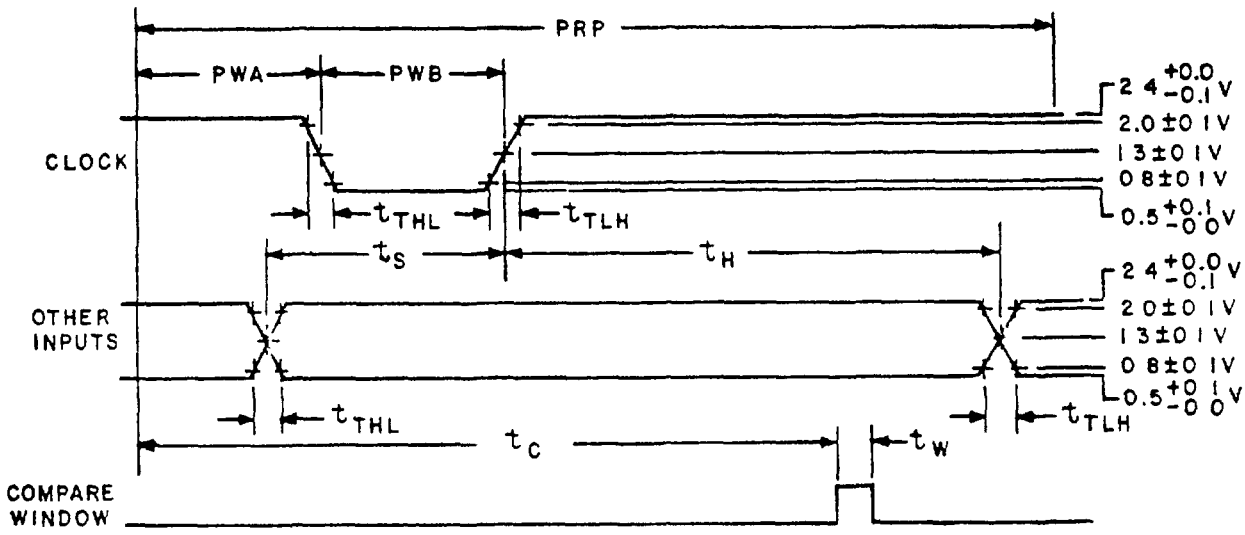
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NOTE Pulse parameters
PRP = 500 ±5 ns.
PWA = 200 ±2 ns.
PWB = 100 ±2 ns.
t_C = 140 ±2 ns
t_W = 10 ±2 ns.
t_S = 300 ±2 ns.
t_H = 6 ±2 ns (C_1).
= 5 ±2 ns ($\overline{RE}$)
= 3 ±2 ns (D0-D3)
= 1 ±2 ns (All otners).
$t_{TLH} = t_{THL}$ = 5 ±5 ns

FIGURE 18. Dynamic functional test (hold time) waveforms for table XIV

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NOTE. Pulse parameters
PRP = 500 ±5 ns
PWA = 100 ±2 ns.
PWB = 100 ±2 ns.
t_C = 380 ±2 ns
t_W = 10 ±2 ns.
t_H = 270 ±2 ns.
t_S = 31 ±2 ns (D0-D3).
200 ±2 ns (All others)

FIGURE 19. Dynamic functional test (data to auxillary register setup time) waveforms for table XIV.

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* TABLE XV. Setup and hold times (all relative to clock low-to-high transition).

From input	CP. Setup time (ns)	Hold time (ns)
$\overline{I0}$-I4	124	0
$\overline{CC}$	80	↓
$\overline{IEN}$	69	↓
C_n	52	↓
C_1	37	5
D(RE=L; I_4-0=0 ⟶ 8 or 10 ⟶ 15)	30	2
D(all other conditions)	72	2
$\overline{RE}$	29	4

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TABLE XVI. Transition time (t_{TLH}) test.

Test condition	Parameter		Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Connect: F to pin	Connect: I to pin	Connect: G to pin	Connect: H to pin	Initial limits 25° 5°C ambient: Min	Initial limits 25° 5°C ambient: Max	Unit	Test figure
---	Setup		Open	12,13, 14,15, 16,17, 31,34, 37	4	Open	Open	Open	Open	Open	---	---	---	20,21
P1	Y0	Transition time (t_{TLH})	14,34, 37	12,13, 31,15, 16,17	Open	6,7,8, 9				30	2	17	n	20,22
P2	Y1									29				
P3	Y2									28				
P4	Y3									27				
P5	$\overline{G}$									26				
P6	C_{n+4}									25				
P7	C_{i+4}									32				
---	Setup		6,7,8, 9	12,14, 15,16, 17,31, 34,37,	4	Open				Open	---	---	---	20,21
P8	$\overline{P}$	Transition time (t_{TLH})	14,16, 34	12,13, 15,17, 31,37	Open	6,7,8, 9				5	2	17	n	20,22
---	Setup		6,7,8, 9	12,14, 15,16, 17,31, 34,37,	4	Open	Open			Open	---	---	---	20,21
P9	$\overline{EMPTY}$	Transition time (t_{TLH})	6,7,8, 9,15, 16	12,14, 17,31, 34	Open	4	37		4	35	2	17	n	20,22
P10	$\overline{FULL}$	Transition time (t_{TLH})	6,7,8, 9,15, 16	12,14, 17,31, 34	Open	4	37		4	36	2	17	n	20,22

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TABLE XVII. Transition time (t_{THL}) tests.

Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Connect F to pin	Connect G to pin	Connect H to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
---	Setup		Open	12,13, 14,15, 16,17, 31,34, 37	4	Open	Open	Open	Open	Open	---	---	---	20, 21
Q1	Y0	Transition time (t_{THL})	14,34, 37	12,13, 31,15, 16,17	Open	6,7,8, 9				30	1	18	ns	20,22
Q2	Y1									29				
Q3	Y2									28				
Q4	Y3									27				
Q5	$\overline{G}$									26				
Q6	C_{n+4}									25				
Q7	C_{i+4}									32				
---	Setup		6,7,8, 9	12,14, 15,16, 17,31, 34,37,	4	Open				Open	---	---	---	20, 21
Q8	$\overline{P}$	Transition time (t_{THL})	14,16, 34	12,13, 15,17, 31,37	Open	6,7,8, 9				5	1	18	ns	20,22
---	Setup		6,7,8, 9	12,14, 15,16, 17,31, 34,37,	4	Open				Open	---	---	---	20, 21
Q9	$\overline{EMPTY}$	Transition time (t_{THL})	6,7,8, 9,15, 16	12,14, 17,31, 34	Open	4	37		4	35	1	18	ns	20,22
Q10	$\overline{FULL}$	Transition time (t_{THL})	6,7,8, 9,15, 16	12,14, 17,31, 34	Open	4	37		4	36	1	18	ns	20,22

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* TABLE XVIII. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect								Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	H to pin	Min	Max		
---	Setup	6,7,8, 9	14,15, 16,17, 12,31, 34,37,	Open	4	Open	Open	Open	Open	---	---	---	20,21
R1	I0→Y0 Propagation delay time (t_{PLH})	6,7,8, 9	14,15, 16,12, 31,34		17,37			37	30	7	70	ns	20,22
---	Setup	6,7,8, 9	14,15, 16,17, 12,31, 34,37,		4			Open	Open	---	---	---	20,21
R2	I0→$\overline{G}$ Propagation delay time (t_{PLH})	Open	12,14, 15,17, 31,34, 6,7,8, 9		16,37			37	26	5	58	ns	20,22
R3	I0→$\overline{P}$ Propagation delay time (t_{PLH})		12,14, 15,17, 31,34, 6,7,8, 9		16,37			37	5	5	58	ns	20,22
R4	I0→C_{n+4} Propagation delay time (t_{PLH})		12,14, 15,17, 31,34, 6,7,8, 9		16,37			37	25	6	65	ns	20,22
---	Setup		6,7,8, 9,12, 14,15, 16,17, 31,34, 37		4			Open	Open	---	---	---	20,21
R5	I1→C_{1+4}, I4=1 Propagation delay time (t_{PLH})	6,7,8, 9,14, 34,37	12,13, 15,16, 31	4	17			17	32	7	78	ns	20,23
---	Setup	6,7,8, 9	12,31, 14,15, 16,17, 34,37	Open	4			Open	Open	---	---	---	20,21

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* TABLE XVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect								Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	H to pin	Min	Max		
R6	$\overline{I1} \rightarrow C_{i+4}$, I4 = 0 Propagation delay time (t_{PLH})	6,7,8, 9,15, 34	12,14, 16,31, 37	4	17	Open	Open	4	32	6	69	ns	20,25
---	Setup	6,7,8, 9	12,13, 14,15, 16,17, 31,34, 37	Open	4			Open	Open	---	---	---	20,21
R7	$\overline{CC} \rightarrow$ Y0 Propagation delay time (t_{PLH})	6,7,8, 9,14, 34	12,31, 15,16, 17		13,37			13	20	5	53	ns	20,22
R8	$\overline{CC} \rightarrow \overline{G}$	6,7,8, 9,14, 34	12,31, 15,16, 17		13,37				26	3	39		
R9	$\overline{CC} \rightarrow C_{n+4}$								25	4	46		
R10	$\overline{CC} \rightarrow C_{i+4}$, I4=1								32	6	61		
---	Setup	6,7,8, 9	14,15, 16,17, 12,31, 34,37		4			Open	Open	---	---	---	20,21
R11	$\overline{CC} \rightarrow \overline{P}$ Propagation delay time (t_{PLH})	6,7,8, 14,16	9,12, 15,17, 31,34, 37		13			13	5	3	39	ns	20,22
---	Setup	6,7,8, 9	12,13, 14,15, 16,17, 31,34, 37		4			Open	Open	---	---	---	20,21
R12	$C_n \rightarrow$ Y0 Propagation delay time (t_{PLH})	6,7,8, 9,14, 16,34	12,13, 15,17, 37		31			31	30	2	29	ns	20,22

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TABLE XVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect								Initial limits 25°+5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	H to pin	Min	Max		
R13	$C_n \rightarrow C_{n+4}$ Propagation delay time (t_{PLH})	6,7,8, 9,14, 16, 34	12,13 15,17, 37	Open	31	Open	Open	31	25	2	21	ns	20,22
R14	$C_n \rightarrow C_{1+4}$, I4=1 Propagation delay time (t_{PLH})	6,7,8, 9,14, 16,34	12,13 15,17, 37		31			31	32	3	36	ns	20,22
---	Setup	6,7,8, 9	12,13 14,15, 16,17, 31,34 37		4			Open	Open	---	---	--	20,21
---	Setup (1)	6,7,8, 9,17, 37,34	12,13, 14,15, 16,31		4			Open	Open	---	---	---	20,21
R15	$C_i \rightarrow C_{1+4}$, I4=0 Propagation delay time (t_{PLH})	6,7,8, 9,17, 37	12,13, 14,15 16,31		34			34	32	2	17	ns	20,22
---	Setup	6,7,8, 9	12,13, 14,15, 16,17 31,34, 37		4			Open	Open	---	---	-	20,21
R16	$C_i \rightarrow C_{1+4}$, I4=1 Propagation delay time (t_{PLH})	6,7,8, 9,14, 37	12,13, 15,16, 17,31		34			34	32	2	17	ns	20,22
---	Setup	6,7,8, 9	12,13, 14,15, 16,17 31,34, 37		4			Open	Open	---	---	---	20,21
R17	CP $\rightarrow$ Y0 Propagation delay time (t_{PLH})	6,7,8, 14,16, 34	12, 14, 17, 31		4			4	30	5	53	ns	20,22

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* TABLE XVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Connect F to pin	Connect G to pin	Connect H to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
R18	CP→$\overline{G}$ Propagation delay time (t_{PLH})	6,7,8, 14,16, 34	12,13, 15,17, 9,31, 37	Open	4	Open	Open	4	26	4	46	ns	20,27
R19	CP→$\overline{P}$								5	4	46		20,24
R20	CP→C_{n+4}								25	5	53		20,27
R21	CP→C_{i+4}, I4=1								32	6	66		20,26
---	Setup	6,7,8, 9	12,14, 15,16, 17,31, 34,37					Open	Open	---	---	---	20,21
R22	CP→C_{i+4}, I4=0 Propagation delay time (t_{PLH})	6,7,8, 9,34, 37	12,31, 14,15, 16,17					4	32	3	38	ns	20,26
---	Setup	6,7,8, 9	12,31, 14,15, 16,17, 34,37					Open	Open	---	---	---	20,21
R23	CP→$\overline{FMPTY}$ Propagation delay time (t_{PLH})	6,7,8, 9,15, 16	12,14, 17,31, 34			37		4	35	4	47	ns	20,28
R24	CP→$\overline{FULL}$ Propagation delay time (t_{PLH})	6,7,8, 9,15, 16	12,14, 17,31, 34			37		4	36	4	47	ns	20,28
---	Setup	Open	12,14, 15,16, 17,31, 34,37			Open		Open	Open	---	---	---	20,21
R25	D0→Y0 Propagation delay time (t_{PLH})	17,37	12,14, 15,16 31,34		9	Open		9	30	4	43	ns	20,22

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* TABLE XVIII Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect								Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	H to pin	Min	Max		
---	Setup	Open	12,13, 14,15, 16,17, 31,34, 37	Open	4	Open	Open	Open	Open	---	---	---	20,21
R26	D0 → $\overline{G}$ Propagation delay time (t_{PLH})	14,34, 37	12,13, 15,16, 17,31		6,7,8, 9			9	26	2	28	ns	20,22
R27	D0 → C_{n+4} Propagation delay time (t_{PLH})	14,34, 37	12,13, 15,16, 17,31		6,7,8, 9			9	25	3	35	ns	20,22
R28	D0 → C_{i+4}, Propagation delay I4=1 time (t_{PLH})	14,34, 37	12,13, 15,16, 17,31		6,7,8, 9			9	32	5	50	ns	20,22
---	Setup	Open	12,13, 14,15, 16,17, 31,34, 37		4			Open	Open	---	---	---	20,21
R29	D0 → $\overline{P}$ Propagation delay time (t_{PLH})	6,7,8, 14,15, 16,37	12,13, 17,31, 34		9			9	5	2	28	ns	20,22

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TABLE XIX. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect								Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	H to pin	Min	Max		
---	Setup	6,7,8,9	14,15,16,17,12,31,34,37,	Open	4	Open	Open	Open	Open	---	---	---	20,21
S1	I0 → Y0 Propagation delay time (t_{PHL})	6,7,8,9	14,15,16,12,31,34		17,37			37	30	7	70	ns	20,22
---	Setup	6,7,8,9	14,15,16,17,12,31,34,37,		4			Open	Open	---	---	---	20,21
S2	I0 → $\overline{G}$ Propagation delay time (t_{PHL})	Open	12,14,15,17,31,34,6,7,8,9		16,37			37	26	5	58	ns	20,22
S3	I0 → $\overline{P}$ Propagation delay time (t_{PHL})		12,14,15,17,31,34,6,7,8,9		16,37			37	5	5	58	ns	20,22
S4	I0 → C_{n+4} Propagation delay time (t_{PHL})		12,14,15,17,31,34,6,7,8,9		16,37			37	25	6	65	ns	20,22
---	Setup		6,7,8,9,12,14,15,16,17,31,34,37		4			Open	Open	---	---	---	20,21
S5	I1 → C_{i+4}, I4=1 Propagation delay time (t_{PHL})	6,7,8,9,14,34,37	12,13,15,16,31	4	17			17	32	7	78	ns	20,23
---	Setup	6,7,8,9	12,31,14,15,16,17,34,37	Open	4			Open	Open	---	---	---	20,21

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TABLE XIV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Connect F to pin	Connect G to pin	Connect H to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
S6	$\overline{I1} \rightarrow C_{1+4}$, I4=0 Propagation delay time (t_{PHL})	6,7,8, 9,15, 34	12,14 16,31, 37	4	17	Open	Open	4	32	6	69	ns	20,25
---	Setup	6,7,8, 9	12,13, 14,15, 16,17, 31,34, 37	Open	4			Open	Open	---	---	---	20,21
S7	$\overline{CC} \rightarrow$ YO Propagation delay time (t_{PHL})	6,7,8, 9,14, 34	12,31, 15,16, 17		13,37			13	30	5	53	ns	20,22
S8	$\overline{CC} \rightarrow \overline{G}$								26	3	39		
S9	$\overline{CC} \rightarrow C_{n+4}$								25	4	46		
S10	$\overline{CC} \rightarrow C_{1+4}$, I4=1								32	6	61		
---	Setup	6,7,8, 9	14,15, 16,17, 12,31, 34,37		4			Open	Open	---	---	---	20,21
S11	$\overline{CC} \rightarrow \overline{P}$ Propagation delay time (t_{PHL})	6,7,8, 14,16	9,12, 15,17, 31,34, 37		13			13	5	3	39	ns	20,22
---	Setup	6,7,8, 9	12,13, 14,15, 16,17, 31,34, 37		4			Open	Open	---	---	---	20,21
S12	$C_n \rightarrow$ YO Propagation delay time (t_{PHL})	6,7,8, 9,14, 16,34	12,13, 15,17, 37		31			31	30	2	29	ns	20,22

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TABLE XIX. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect								Initial limits 25°+5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	H to pin	Min	Max		
S13	$C_n \rightarrow C_{n+4}$ Propagation delay time (t_{PHL})	6,7,8, 9,14, 16,34	12,13, 15,17, 37	Open	31	Open	Open	31	25	2	21	ns	20,22
S14	$C_n \rightarrow C_{i+4}$, I4=1 Propagation delay time (t_{PHL})	6,7,8, 9,14, 16,34	12,13, 15,17, 37		31			31	32	3	36	ns	20,22
---	Setup	6,7,8, 9	12,13, 14,15, 16,17, 31,34, 37		4			Open	Open	---	---	---	20,21
---	Setup (A)	6,7,8, 9,17, 37,34	12,13, 14,15, 16,31		4			Open	Open	---	---	---	20,21
S15	$C_i \rightarrow C_{i+4}$, I4=0 Propagation delay time (t_{PHL})	6,7,8, 9,17, 37	12,13, 14,15, 16,31		34			34	32	2	17	ns	20,22
---	Setup	6,7,8, 9	12,13, 14,15, 16,17, 31,34, 37		4			Open	Open	---	---	---	20,21
S16	$C_i \rightarrow \bar{C}_{i+4}$, I4=1 Propagation delay time (t_{PHL})	6,7,8, 9,14, 37	12,13, 15,16, 17,31		34			34	32	2	17	ns	20,22
---	Setup	6,7,8, 9	12,13, 14,15, 16,17, 31,34, 37		4			Open	Open	---	---	---	20,21
S17	CP → YO Propagation delay time (t_{PHL})	6,7,8, 14,16, 34	12,13, 15,17, 9, 31, 37		4			4	30	5	59	ns	20,24

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* TABLE XIX. Propagation delay time (t_{PHL}) test - Continued.

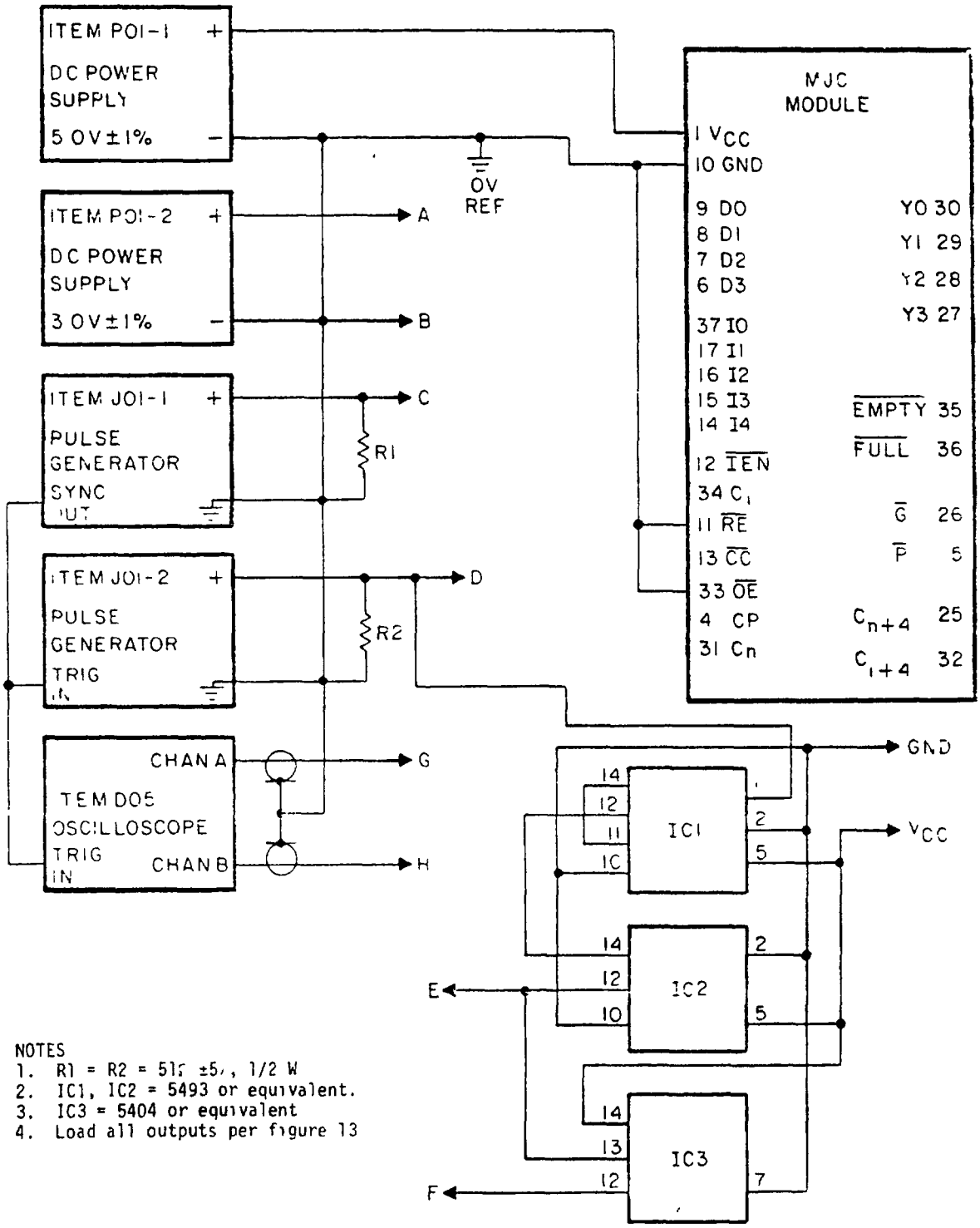
Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Connect F to pin	Connect G to pin	Connect H to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
S18	CP→ $\overline{G}$ Propagation delay time (t_{PHL})	6,7,8, 14,16, 34	12,13, 15,17, 9,31, 37	Open	4	Open	Open	4	26	4	46	ns	20,27
S19	CP→ $\overline{P}$								5	4	46		20,24
S20	CP→ C_{n+4}								25	5	53		20,27
S21	CP→ C_{i+4}, I4=1								32	6	66		20,26
---	Setup	6,7,8, 9	12,14, 15,16, 17,31, 34,37					Open	Open	---	---	---	20,21
S22	CP→ C_{i+4}, I4=0 Propagation delay time (t_{PHL})	6,7,8, 9,34, 37	12,31, 14,15, 16,17					4	32	3	38	ns	20,26
---	Setup	6,7,8, 9	12,31, 14,15, 16,17, 34,37					Open	Open	---	---	---	20,21
S23	CP→ $\overline{EMPTY}$ Propagation delay time (t_{PHL})	6,7,8, 9,15, 16	12,14, 17,31, 34			37		4	35	4	47	ns	20,28
S24	CP→ $\overline{FULL}$ Propagation delay time (t_{PHL})	6,7,8, 9,15, 16	12,14, 17,31, 34			37		4	36	4	47	ns	20,28
---	Setup	Open	12,14, 15,16, 17,31, 34,37			Open		Open	Open	---	---	---	20,21
S25	DO→YO Propagation delay time (t_{PHL})	17,37	12,14, 15,16 31,34		9	Open		9	30	4	43	ns	20,22

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* TABLE XIX. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect								Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	H to pin	Min	Max		
---	Setup	Open	12,13, 14,15, 16,17, 31,34, 37	Open	4	Open	Open	Open	Open	---	---	---	20,21
S26	DO→$\overline{G}$ Propagation delay time (t_{PHL})	14,34, 37	12,13, 15,16, 17,31		6,7,8, 9			9	26	2	28	ns	20,22
S27	DO→C_{n+4} Propagation delay time (t_{PHL})	14,34, 37	12,13, 15,16, 17,31		6,7,8, 9			9	25	3	35	ns	20,22
S28	DO→C_{i+4}, I4=1 Propagation delay time (t_{PHL})	14,34, 37	12,13, 15,16, 17,31		6,7,8, 9			9	32	5	50	ns	20,22
---	Setup	Open	12,13, 14,15, 16,17, 31,34, 37		4			Open	Open	---	---	---	20,21
S29	DO→$\overline{P}$ Propagation delay time (t_{PHL})	6,7,8, 14,15, 16,37	12,13, 17,31, 34		9			9	5	2	28	ns	20,22

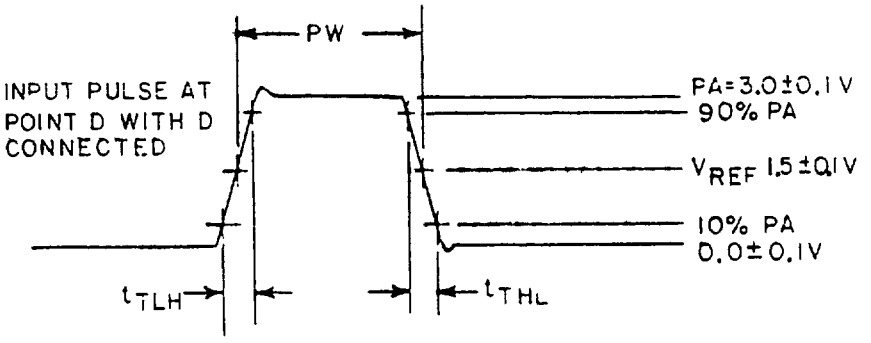
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NOTES
1. R1 = R2 = 51Ω ±5%, 1/2 W
2. IC1, IC2 = 5493 or equivalent.
3. IC3 = 5404 or equivalent
4. Load all outputs per figure 13

FIGURE 20 Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XVI through XIX.

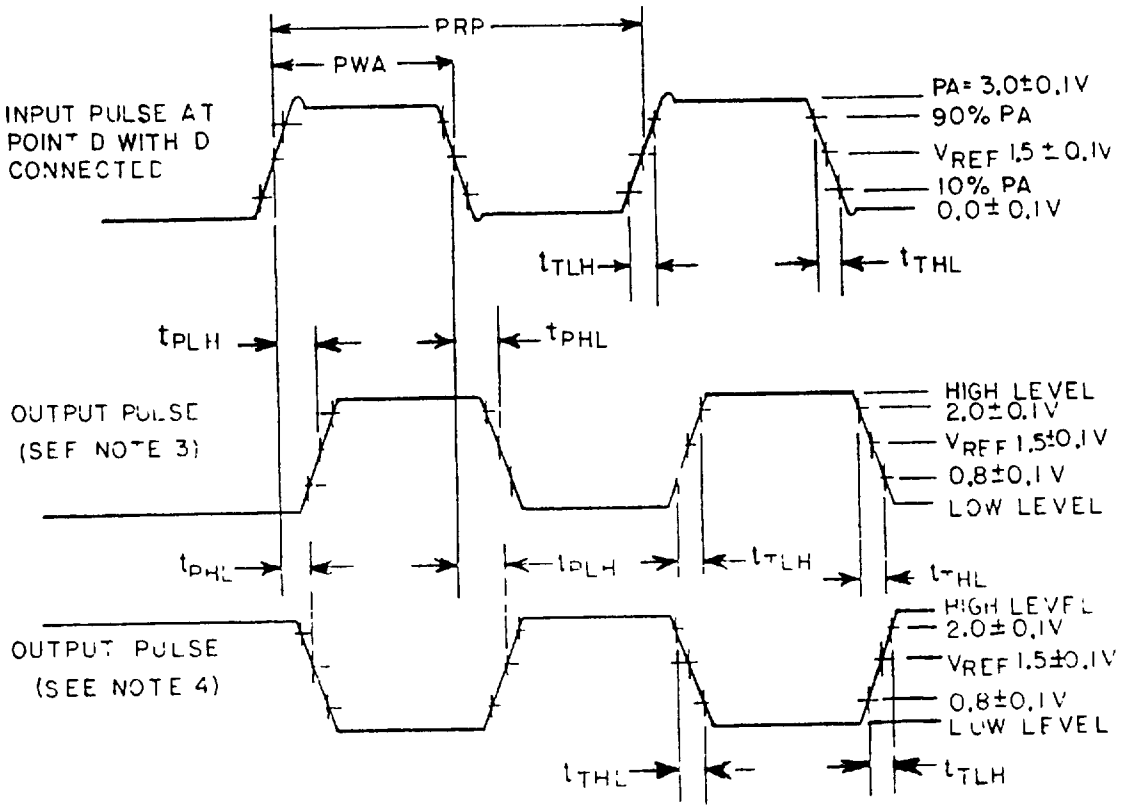
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NOTES
1. Input pulse parameters
   PW = 500 ±50 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Place item J01-2 in single pulse mode for setup conditions
3. For setup A, immediately prior to performing test conditions R15 and S15, clock the module 15 times. After setup (A), insure that the clock input, pin 4, remains low. During setup A, monitor pin 32. It shall go high on the 15th clock pulse

FIGURE 21. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveform for tables XVI through XIX.

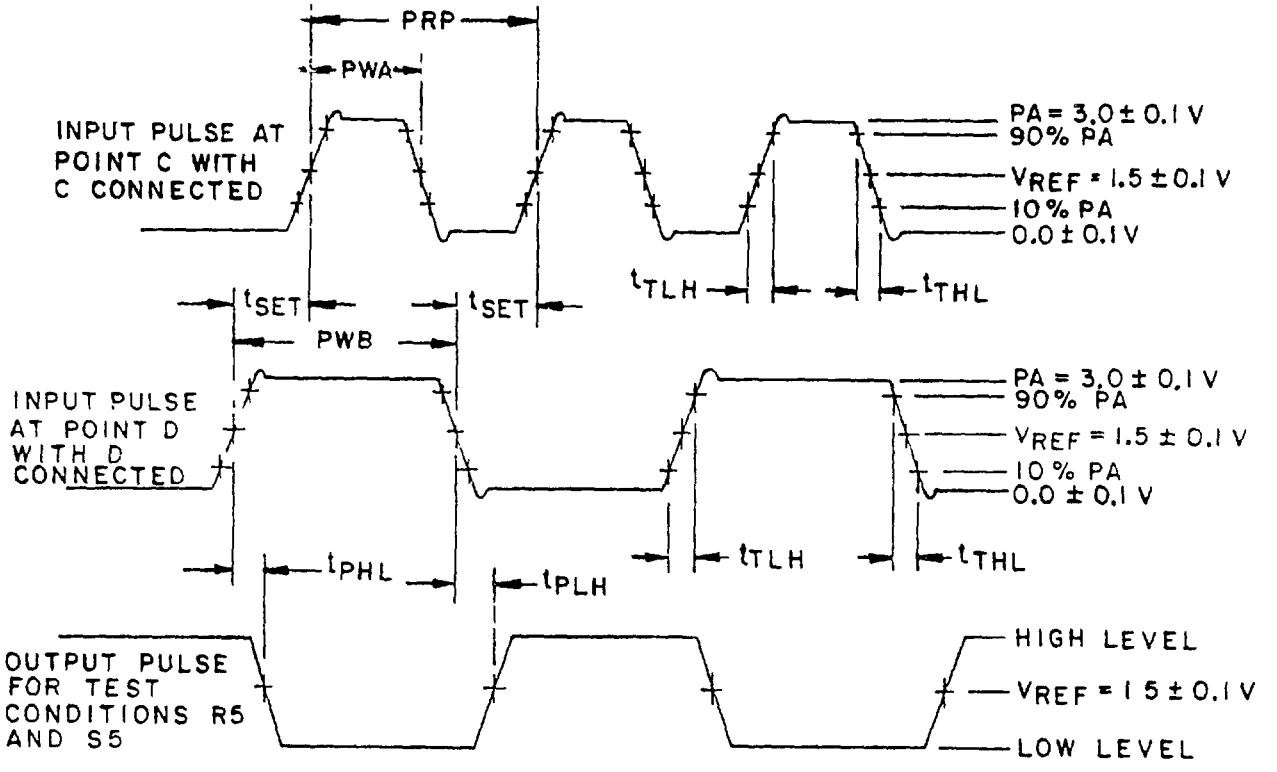
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NOTES
1. Input pulse parameters
   PWA = 500 ±50 ns
   PRP = 1,000 ±100 ns
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
2. Before making the specified measurements, verify the output low level is ≤ 0.5 V and the output high level is ≥ 2.4 V
3. Output waveform for test conditions P1-P4, P6, P7, Q1-Q4, Q6, Q7, R1-R3, R8, R13, R15, R16, R25, R27, R28, S1-S3, S8, S13, S15, S16, S25, S27, S28
4. Output waveform for test conditions P5, P8, Q5, Q8, R4, R7, R9-R12, R14, R26, R29, S4, S7, S9-S12, S14, S26, S29

FIGURE 22. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XVI through XIX.

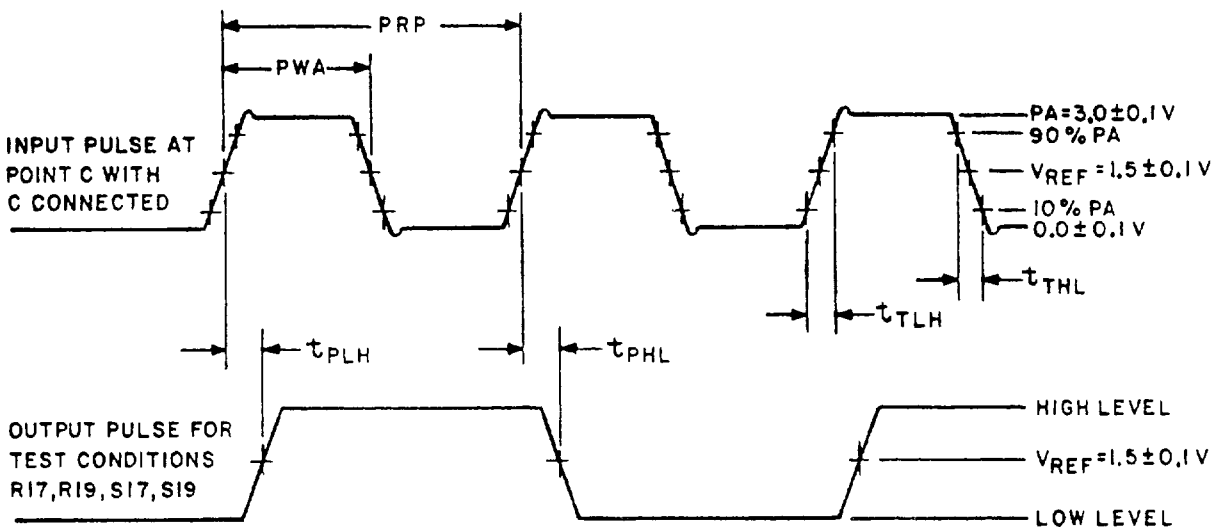
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NOTES
1. Input pulse parameters:
   PWA = 250 ±25 ns, double pulse mode.
   PRP = 500±50 ns.
   PWB = 500 ±50 ns.
   t_{SET} = 124 +5, -0 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns
2. Before making the specified measurement, verify the output low level is $\leq$ 0.5 V and the output high level is $\geq$ 2.4 V.

FIGURE 23. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XVIII and XIX.

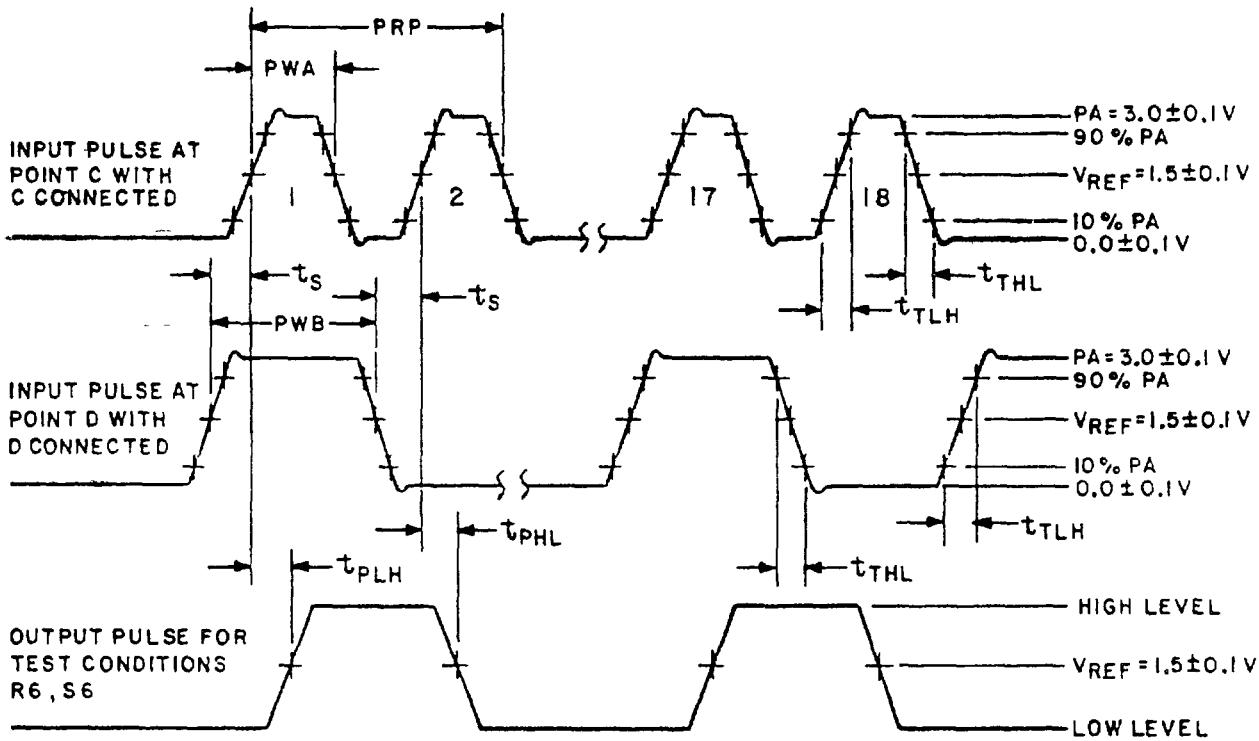
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NOTES
1. Input pulse parameters
   PWA = 250 ±25 ns.
   PRP = 500 ±50 ns
   $t_{TLH} = t_{THL}$ = 10 ±2 ns
2. Before making the specified measurement, verify the output low level is ≤ 0.5 V and the output high level is ≥ 2.4 V

FIGURE 24. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XVIII and XIX.

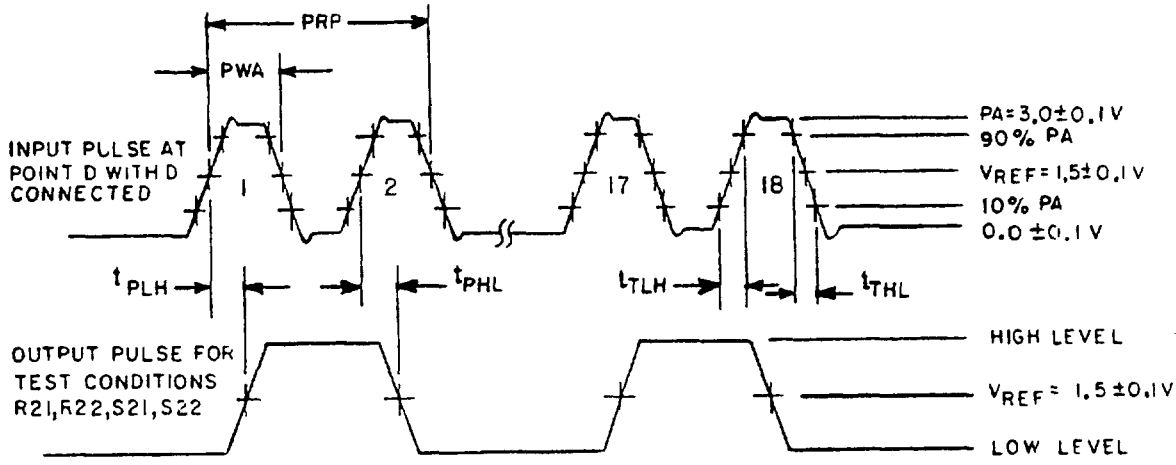
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NOTES
1. Input pulse parameters
   PWA = 250 ±25 ns, double pulse mode.
   PRP = 500 ±50 ns.
   PWB = 500 ±50 ns.
   t_S = 124 +5, -0 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. The output for test conditions R6 and S6 will be low for 15 clock pulses.
3. Before making the specified measurements, verify the output low level is ≤ 0.5 V and the output high level is ≥ 2.4 V.

FIGURE 25. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XVIII and XIX.

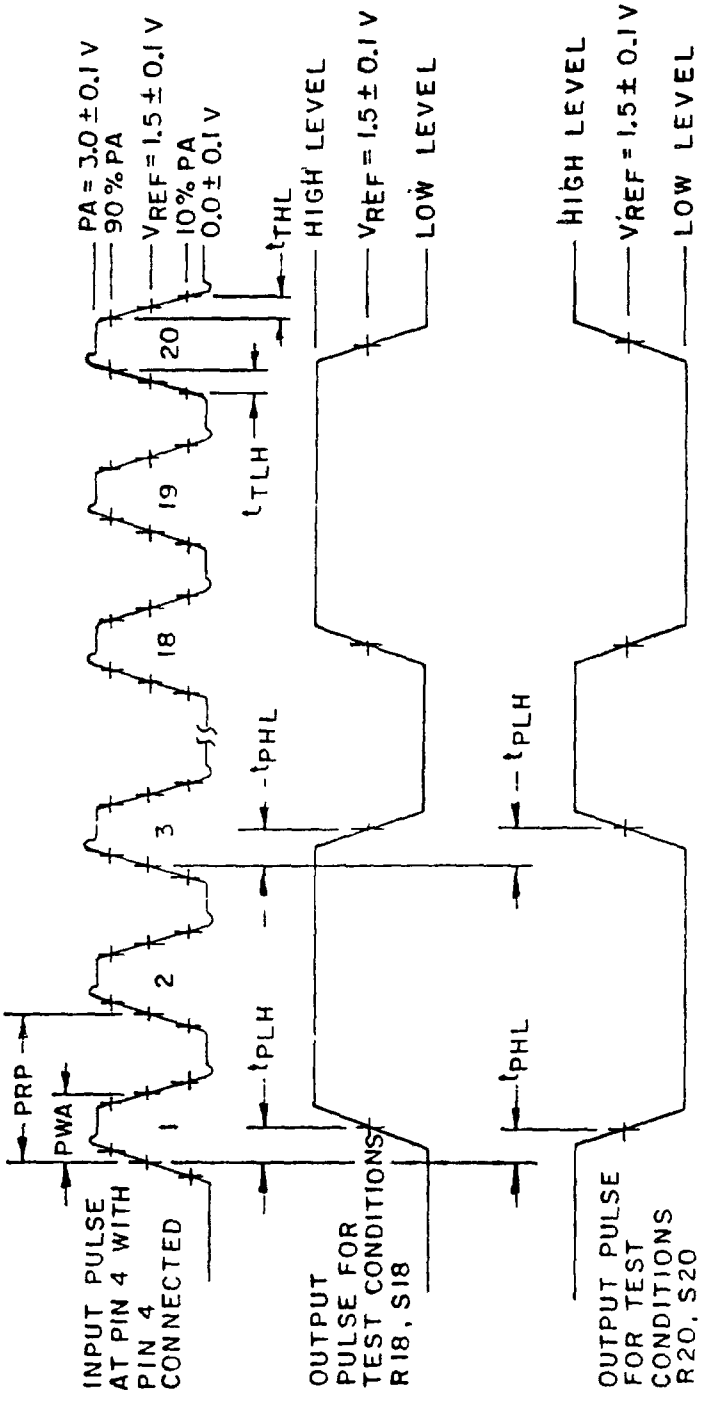
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NOTES
1. Input pulse parameters
   PWA = 250 ±25 ns
   PRP = 500 ±50 ns.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns
2. The output for test conditions R21, R22, S21 and S22 will be low for 15 clock pulses
3. Before making the specified measurements, verify the output low level is ≤ 0.5 V and the output high level is ≥ 2.4 V.

FIGURE 26 Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XVIII and XIX.

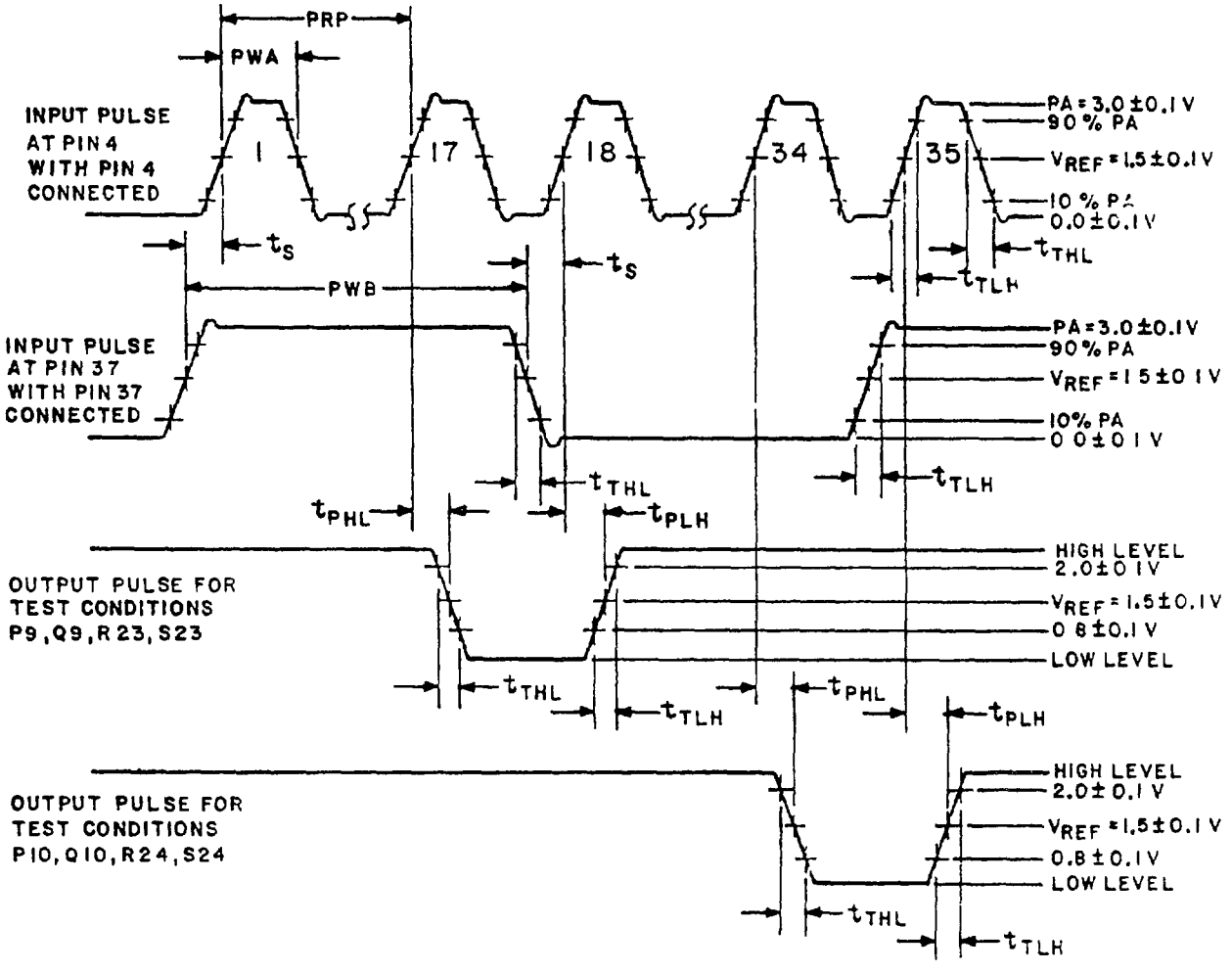
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NOTES:
1. Input pulse parameters:
   PWA = 250 ±25 ns.
   PRP = 500 ±50 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns
2. The output for tests R18 and S18 will be low for 15 clock pulses. The output for tests R20 and S20 will be high for 15 clock pulses.
3. Before making the specified measurement, verify the output low level is ≤ 0.5 V and the output high level is ≥ 2.4 V

FIGURE 27. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XVIII and XIX.

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NOTES
1. Input pulse parameters
   PWA = 50 ±5 ns
   PRP = 100 ±10 ns.
   PWB = 1.7 µs +170 ns
   t_S = 124 +5, -0 ns
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Before making the specified measurement, verify the output low level is ≤ 0.5 V and the output high level is ≥ 2.4 V.

FIGURE 28. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XVI through XIX.

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TABLE XX. Disable recovery time (low-level to high-impedance) test.

Test condition	Parameter	Connect					Initial limits 25°+5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	F to pin	Min	Max		
---	Setup	Open	14, 15, 16, 17, 33, 31, 34, 37	3	Open	Open	---	---	---	29, 30
T1	$\overline{OE}$ to Y0 Disable recovery time (t_{PLZ})		14, 15, 16, 17, 31, 34, 37	33	33	30	5	55	ns	
T2	$\overline{OE}$ to Y1					29				
T3	$\overline{OE}$ to Y2					28				
T4	$\overline{OE}$ to Y3					27				

TABLE XXI. Enable recovery time (high-impedance to low-level) test.

Test condition	Parameter	Connect					Initial limits 25°+5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
U1	$\overline{OE}$ to Y0 Enable recovery time (t_{PZL})	Open	14, 15, 16, 17, 31, 34, 37	33	33	30	5	50	ns	29, 30
U2	$\overline{OE}$ to Y1					29				
U3	$\overline{OE}$ to Y2					28				
U4	$\overline{OE}$ to Y3					27				

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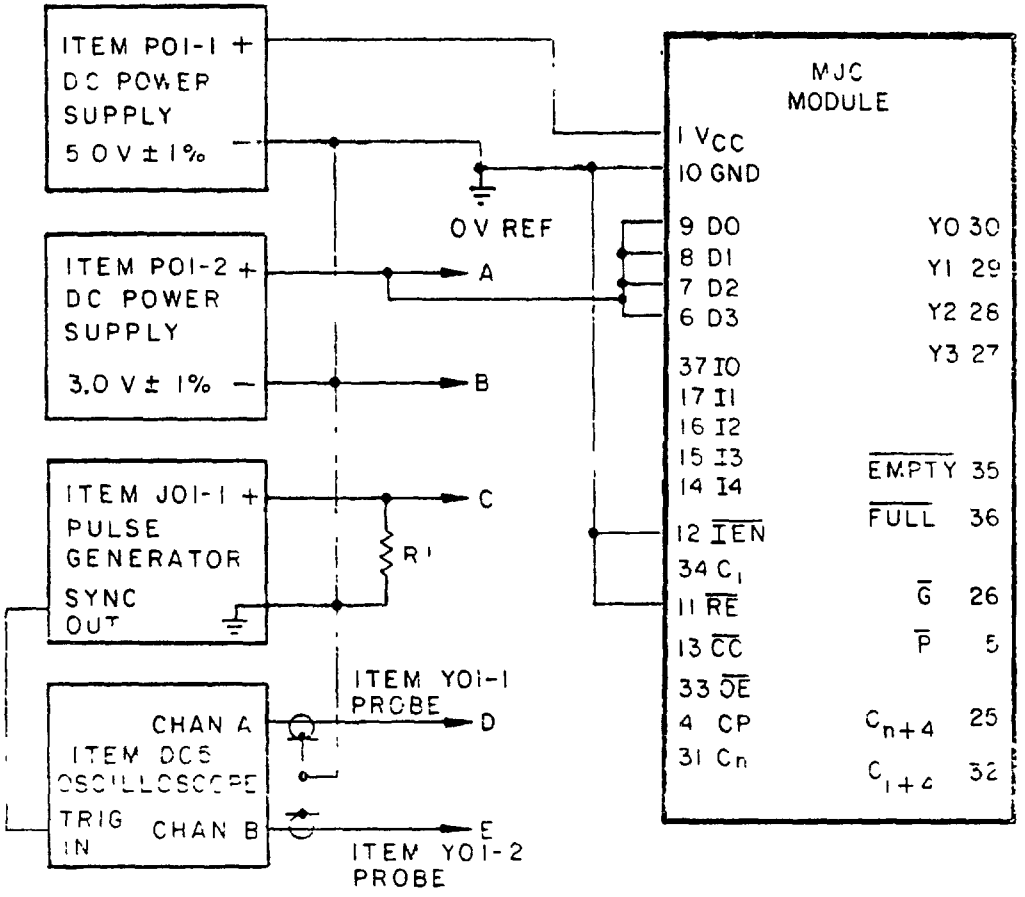
TABLE XXII. Disable recovery time (high-level to high-impedance) test.

Test condition	Parameter	Connect					Initial limits 25°+5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
V1	$\overline{OE}$ to Y0 Disable recovery time (t_{PHZ})	17,37	14,15, 16,31, 34	33	33	30	5	55	ns	29,30
V2	$\overline{OE}$ to Y1					29				
V3	$\overline{OE}$ to Y2					28				
V4	$\overline{OE}$ to Y3					27				

TABLE XXIII. Enable recovery time (high-impedance to high-level) test.

Test condition	Parameter	Connect					Initial limits 25°+5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
W1	$\overline{OE}$ to Y0 Enable recovery time (t_{PZH})	17,37	14,15, 16,31, 34	33	33	30	5	50	ns	29,30
W2	$\overline{OE}$ to Y1					29				
W3	$\overline{OE}$ to Y2					28				
W4	$\overline{OE}$ to Y3					27				

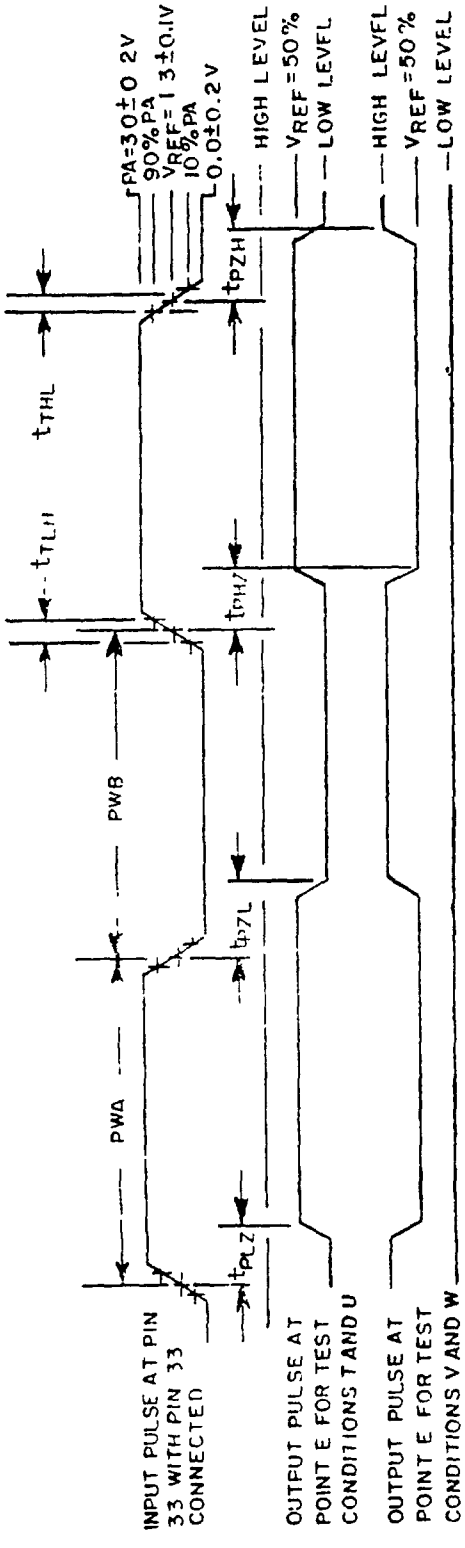
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NOTES.
1. R1 = 51 ±5%, 1/2 W.
2 Load Y outputs per figure 13, note 1.

FIGURE 29. Disable recovery time (low-level and high-level to high impedance) and enable recovery time (high impedance to low-level and high-level) test circuit for tables XX through XXIII.

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NOTES:
1. Input pulse parameters:
   PWA = PWB = 500 ±50 ns.
   $t_{TLH} - t_{THL}$ = 10 ±2 ns.
2. Before making specified measurements, verify that the output low level is ≤ 0.5 V and output high level is ≥ 2.4 V.

*FIGURE 30. Disable recovery time (low-level and high-level to high impedance) and enable recovery time (high impedance to low-level and high-level) test circuit waveforms for tables XX through XXIII.

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TABLE XXJV Output crosstalk (low-level) test.

Test condition	Parameter		Connect					Initial limits 25°+5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
---	Setup		Open	14, 15, 16, 17, 31, 34, 37	4	Open	Open	---	---	---	31, 32
X1	C_{1+4}	Output crosstalk, low level	14, 34, 37	9, 15, 16, 17, 31	Open	6, 7, 8	32	0	500	mV	
X2	Y0			9, 15, 16, 17, 31		6, 7, 8	30				
X3	Y1			8, 15, 16, 17, 31		6, 7, 9	29				
X4	Y2			7, 15, 16, 17, 31		6, 8, 9	28				
X5	Y3			6, 15, 16, 17, 31		7, 8, 9	27				
---	Setup		15, 16, 37, 34	14, 17, 31	4	Open	Open	---	---	---	
X6	$\overline{\text{EMPTY}}$	Output crosstalk, low level	14, 34, 37	15, 16, 17, 31	Open	6, 7, 8, 9	35	0	500	mV	
---	Setup		6, 7, 8, 9, 15, 16, 34	14, 17, 31, 37	4	Open	Open	---	---	---	

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TABLE XXIV. Output crosstalk (low-level) test - Continued.

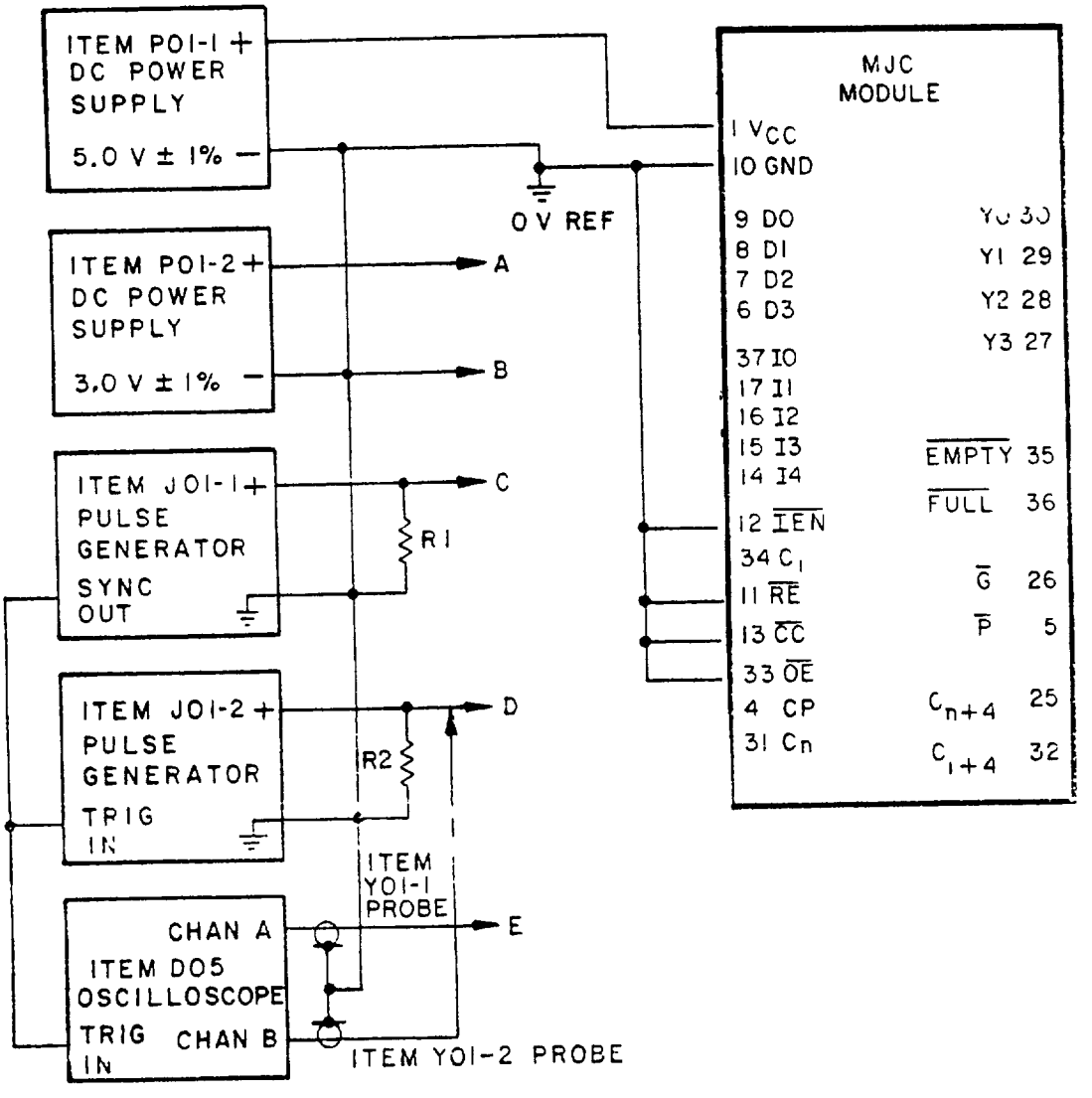
Test condition	Parameter	Connect					Initial limits 25°+5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
X7	$\overline{\text{FULL}}$ Output crosstalk, low level	14,34, 37	15,16, 17,31	Open	6,7,8, 9	36	0	500	mV	31,32
---	Setup	Open	14,15, 16,17, 31,34, 37	4	Open	Open	---	---	---	
X8	$\overline{\text{G}}$ Output crosstalk, low level	9,14, 34,37	15,16, 17,31	Open	6,7,8,	26	0	500	mV	
---	Setup	Open	14,15, 16,17, 31,34, 37	4	Open	Open	---	---	---	
X9	$\overline{\text{P}}$ Output crosstalk, low level	14,34, 37	15,16, 17,31	Open	6,7,8, 9	5	0	500	mV	

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TABLE XXI. Output crosstalk (high level) test

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
---	Setup	Open	14, 15, 16, 17, 31, 34, 37	4	Open	Open	---	---	---	31, 32
Y1	C_{n+4} Output crosstalk, high level	9, 14, 34, 37	15, 16, 17, 31,	Open	6, 7, 8	25	0	500	mV	
Y2	Y0	9, 14, 34, 37			6, 7, 8	30				
Y3	Y1	8, 14, 34, 37			6, 7, 9	29				
Y4	Y2	7, 14, 34, 37			6, 8, 9	28				
Y5	Y3 Output crosstalk, high level	6, 14, 34, 37			7, 8, 9	27				
---	Setup	15, 16, 34, 37	14, 17, 31	4	Open	Open	---	---	---	
Y6	$\overline{FULL}$ Output crosstalk, high level	14, 34, 37	15, 16, 17, 31	Open	6, 7, 8, 9	36	0	500	mV	
---	Setup	6, 7, 8, 9, 15, 16, 34	14, 17, 31, 37	4	Open	Open	---	---	---	
Y7	$\overline{EMPTY}$ Output crosstalk, high level	14, 34, 37	15, 16, 17, 31	Open	6, 7, 8, 9	35	0	500	mV	
---	Setup	6, 7, 8, 9	12, 13, 14, 15, 16, 17, 31, 34, 37	4	Open	Open	---	---	---	
Y8	$\overline{G}$ Output crosstalk, high level	6, 7, 8, 9, 14, 16, 34	12, 13, 15, 17, 37	Open	31	26	0	500	mV	

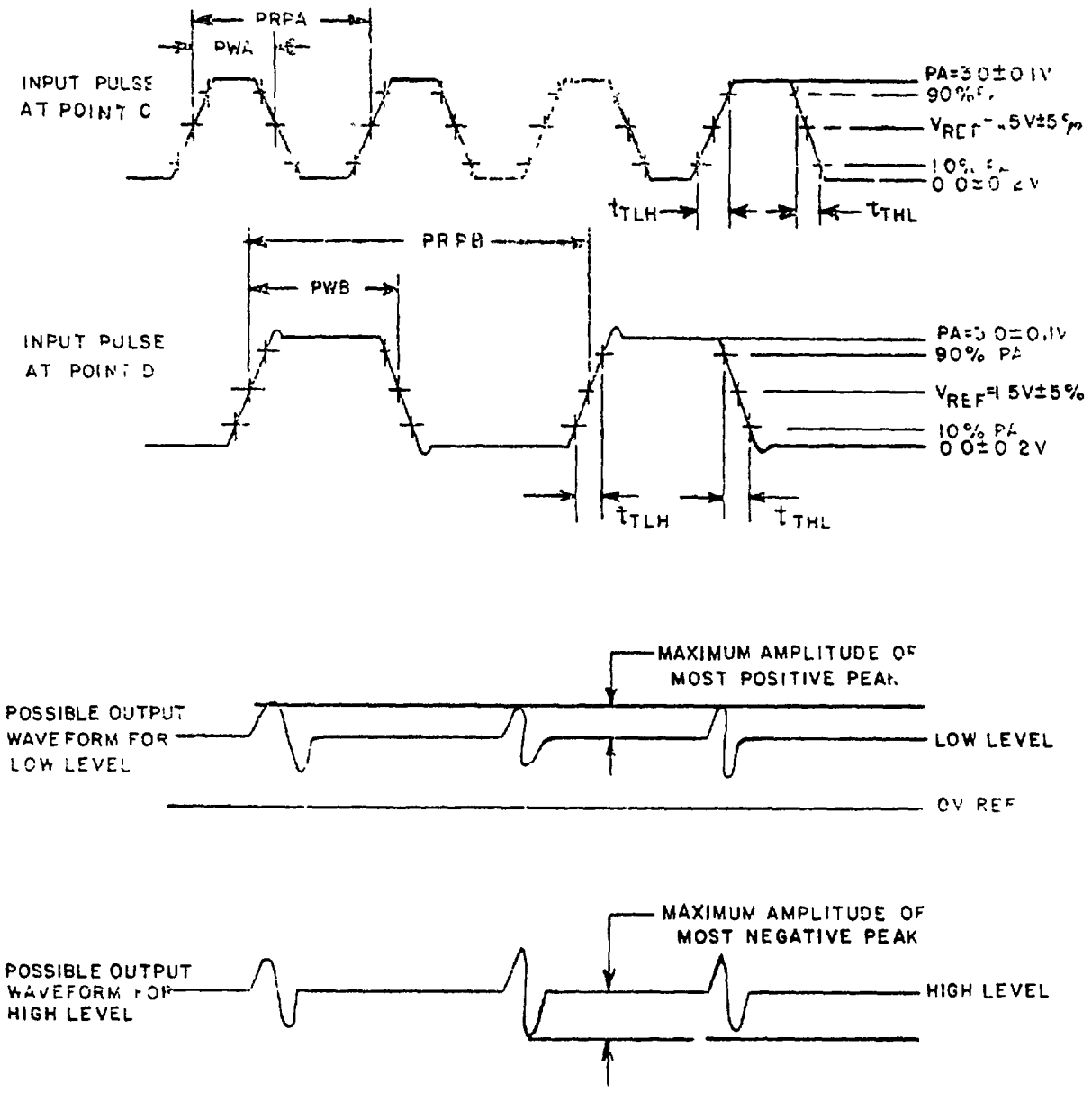
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load all outputs per figure 13

FIGURE 31. Output crosstalk (low-level and high-level) test circuit for tables XXIV and XXV.

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NOTE. Input pulse parameters
PWA = 250 ±25 ns.
PRPA = 500 ±50 ns.
PWB = 500 ±50 ns.
PRPB = 1,000 ±100 ns
$t_{TLH} = t_{THL}$ = 15 ±3 ns.

*FIGURE 32 Output crosstalk (low-level and high-level) test circuit waveforms for tables XXIV and XXV

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TABLE XXVI. Power supply transient amplitude test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Z	Power supply transient amplitude	14,34,37	15,16,17,31	6,7,8,9	0	400	mV	33,34

TABLE XXVII Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
AA	Power supply filter capacitance	1	10	3.15	5.85	µF	35

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TABLF XXVIII. Input capacitance test.

Test condition	Parameter	Connect		Initial limits 25°+5°C ambient		Unit	Test figure
		C to pin	D to pin	Min	Max		
BB1	$\overline{DO}$ Input capacitance	9	10	6	55	pF	35
BB2	$\overline{D1}$	8					
BB3	$\overline{D2}$	7					
BB4	$\overline{D3}$	6					
BB5	$\overline{IO}$	37		2	25		
BB6	$\overline{I1}$	17					
BB7	$\overline{I2}$	16					
BB8	$\overline{I3}$	15					
BB9	$\overline{I4}$	14					
BB10	$\overline{IEN}$	12					
BB11	C_i	34					
BB12	$\overline{RE}$	11		4	40		
BB13	$\overline{CC}$	13		2	25		
BB14	$\overline{OE}$	33					
BB15	CP	4					
BB16	C_n	31					

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TABLE XXIX. Output capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		C to pin	D to pin	Min	Max		
CC1	YO Output capacitance	30	10	2	30	pF	35
CC2	Y1	29					
CC3	Y2	28					
CC4	Y3	27					
CC5	$\overline{\text{EMPTY}}$	35					
CC6	$\overline{\text{FULI}}$	36					
CC7	$\overline{G}$	26					
CC8	$\overline{P}$	5					
CC9	C_{n+4}	25					
CC10	C_{1+4}	32					

* TABLE XXX. Enable recovery time (I4=0 to Y, t_{PZL}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
DD1	I4-IO to Y Enable recovery time (t_{PZL})	17,37	6,7,8,9	14,15,16	30	8	80	ns	3[illegible]
DD2					29				
DD3					28				
DD4					27				

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* TABLE XXXI. Disable recovery time (I4=0 to Y, t_{PLZ}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
EE1	I4-IO to Y Disable recovery time (t_{PLZ})	17,37	6,7,8,9	14,15,16	30	6	60	ns	36,37
EE2	↓	↓	↓	↓	29	↓	↓	↓	↓
EE3					28				
EE4					27				

* TABLE XXXII. Disable recovery time (I4=0 to Y, t_{PHZ}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
FF1	I4-IO to Y Disable recovery time (t_{PHZ})	6,7,8,9,17, 37	Open	14,15,16	30	6	60	ns	36,37
FF2	↓	↓	↓	↓	29	↓	↓	↓	↓
FF3					28				
FF4					27				

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TABLE XXXIII. Enable recovery time (I4-0 to Y, t_{PZH}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
GG1	I4-I0 to Y Disable recovery time (t_{PZH})	6,7,8,9,17, 37	Open	14,15,16	30	8	80	ns	36,37
GG2					29				
GG3					28				
GG4					27				

TABLE XXXIV. Enable recovery time ($\overline{CC}$ to Y, t_{PZL}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
---	Setup	Open	14,15,16,17, 37,12,34	4	Open	---	---	---	38,39
HH1	$\overline{CC}$ to Y Enable recovery time (t_{PZL})	14,15,16, 17,37	12,34	13	30	6	60	ns	
HH2					29				
HH3					28				
HH4					27				

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TABLE XXXV. Disable recovery time ($\overline{CC}$ to Y, t_{PLZ}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
---	Setup	Open	14,15,16,17, 37,12,34	4	Open	---	---	---	38,39
JJ1	$\overline{CC}$ to Y Disable recovery time (t_{PLZ})	14,15,16, 17,37	12,34	13	30	5	50	ns	
JJ2					29				
JJ3					28				
JJ4					27				

TABLE XXXVI. Enable recovery time ($\overline{CC}$ to Y, t_{PZH}) test.

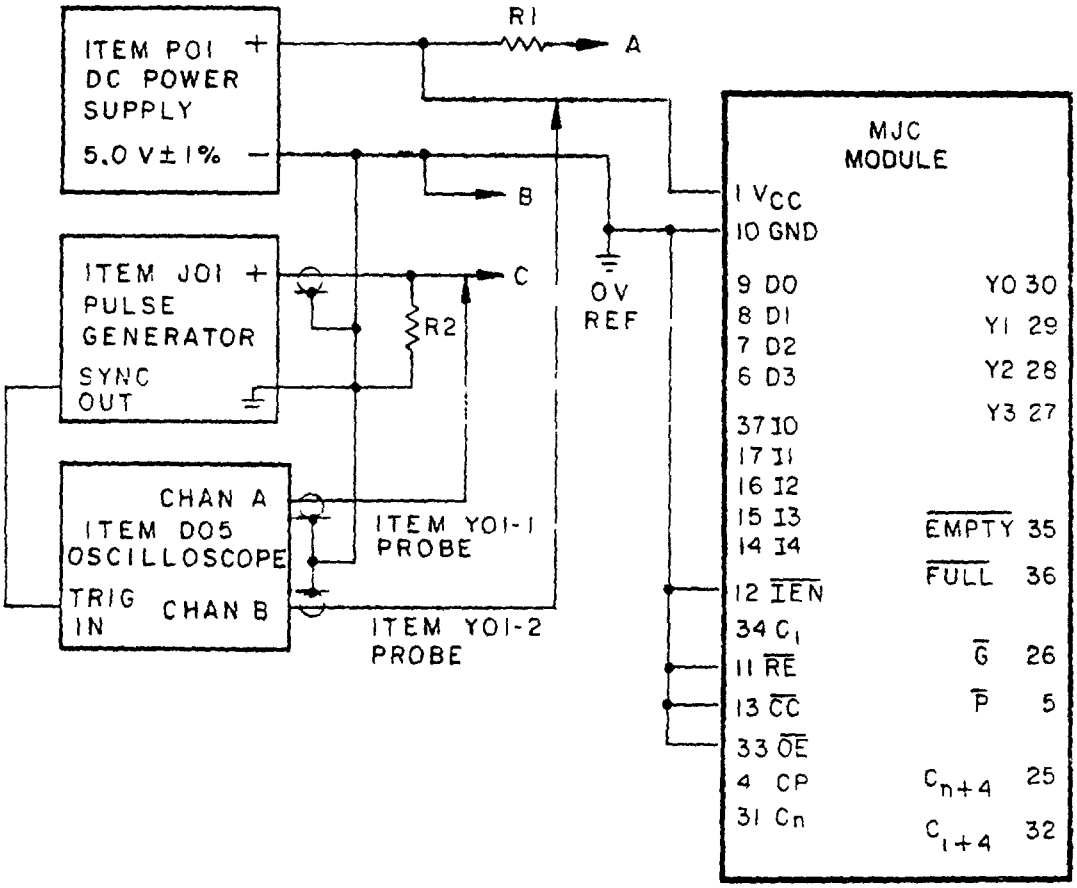
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
---	Setup	6,7,8,9, 14,37	15,16,17,12, 13,34	4	Open	---	---	---	38,39
KK1	$\overline{CC}$ to Y Enable recovery time (t_{PZH})	14,15,16, 17,37	12,34	13	30	6	60	ns	
KK2					29				
KK3					28				
KK4					27				

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TABLE XXXVII Disable recovery time ($\overline{CC}$ to Y, t_{PHZ}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
---	Setup	6,7,8,9, 14,37	15,16,17,12, 13,34	4	Open	---	---	---	39,39
LL1	$\overline{CC}$ to Y Disable recovery time (t_{PHZ})	14,15,16, 17,37	12,34	13	30	5	50	ns	
LL2					29				
LL3					28				
LL4					27				

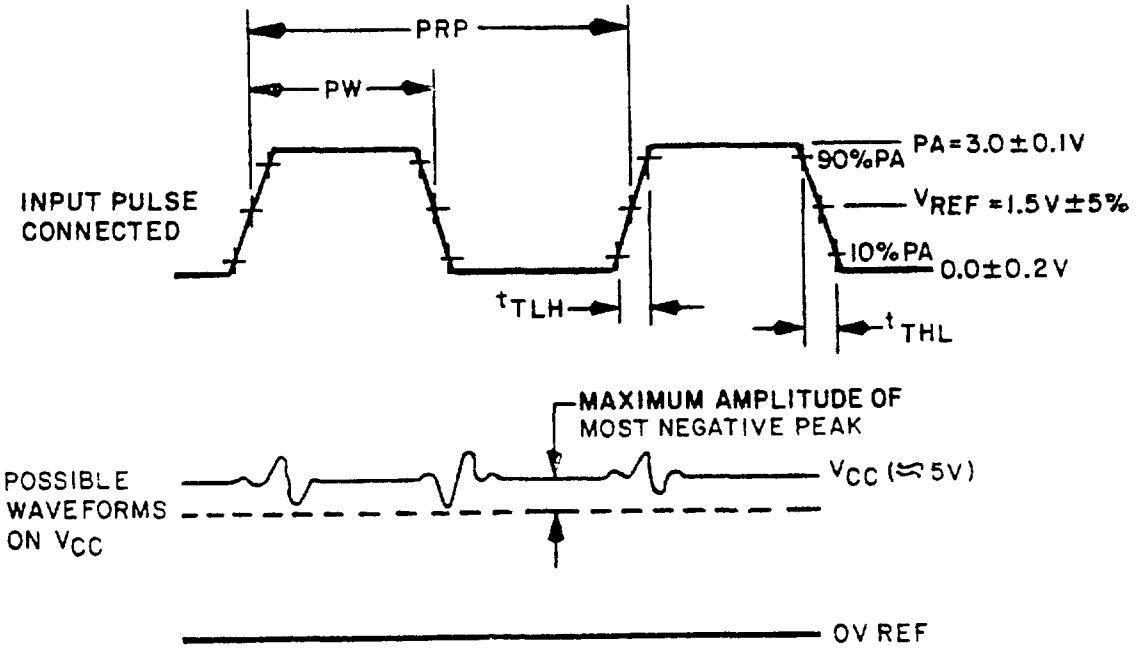
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NOTES.
1. R1 = 1 kΩ ±5%, 1/4 W.
2. R2 = 51Ω ±5%, 1/2 W
3 Load all outputs per figure 13.

FIGURE 33. Power supply transient amplitude test circuit for table XXVI.

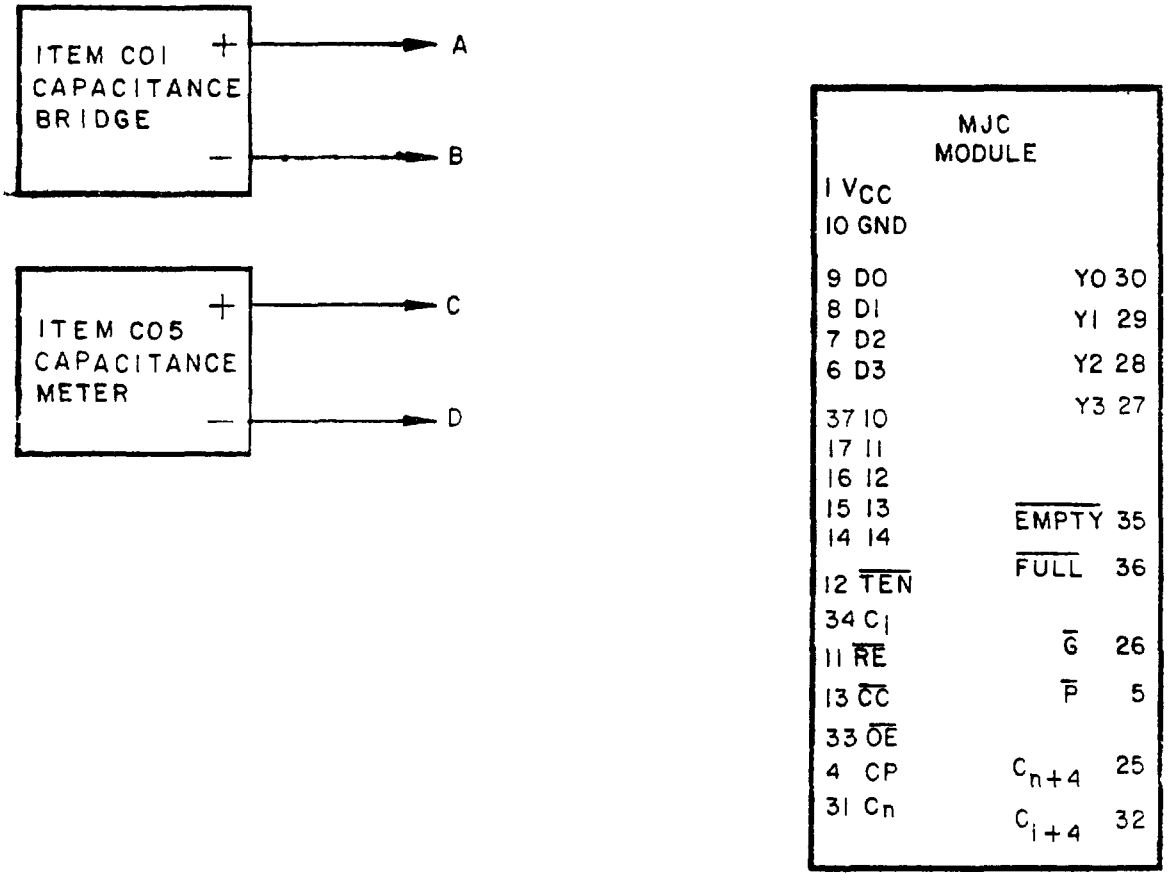
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NOTE: Input pulse setup.
PW = 500 ±50 ns
PRP = 1,000 ±100 ns.
$t_{TLH} = t_{THL} = 10 \pm 2$ ns.

FIGURE 34. Power supply transient amplitude test circuit waveforms for table XXVI.

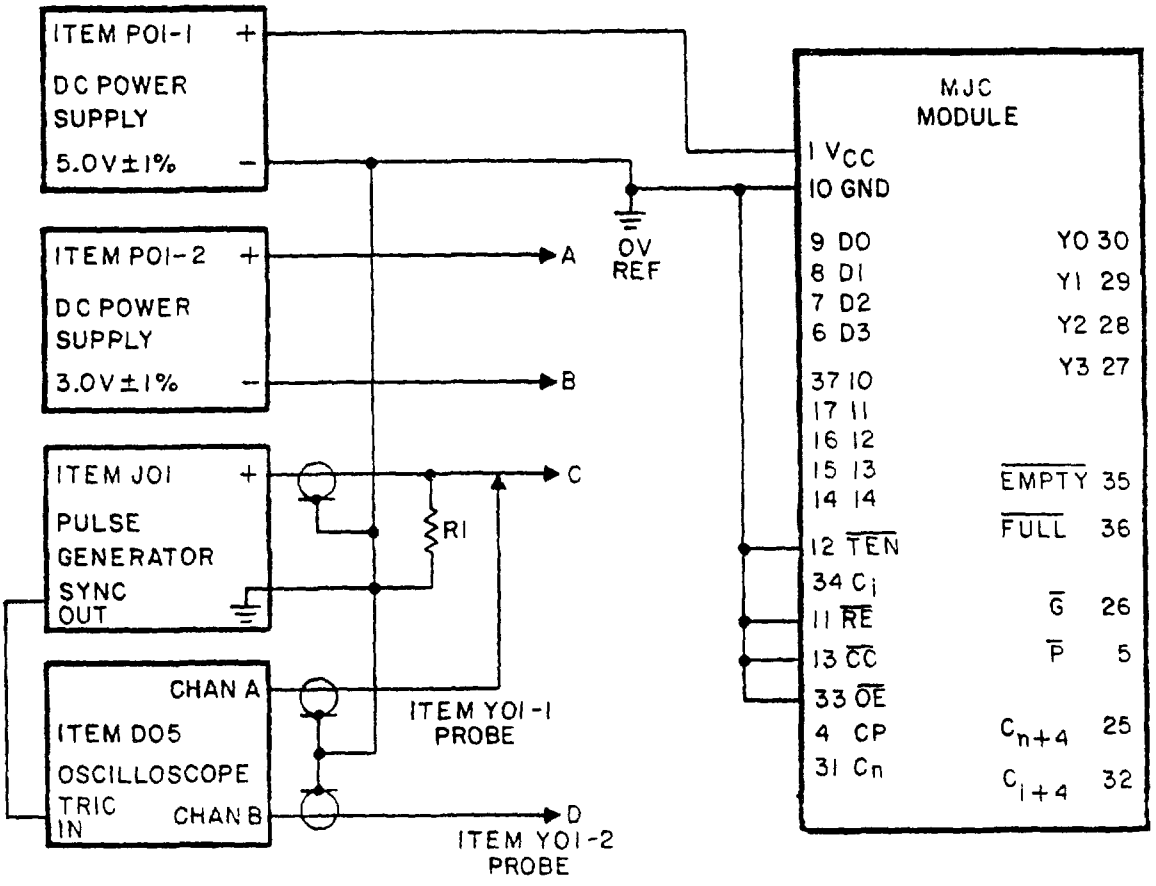
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NOTES
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the reading.
3. Capacitance bridge (C01) setup:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V maximum.

*FIGURE 35 Power supply filter, input and output capacitance test circuit for tables XXVII, XXVIII, and XXIX.

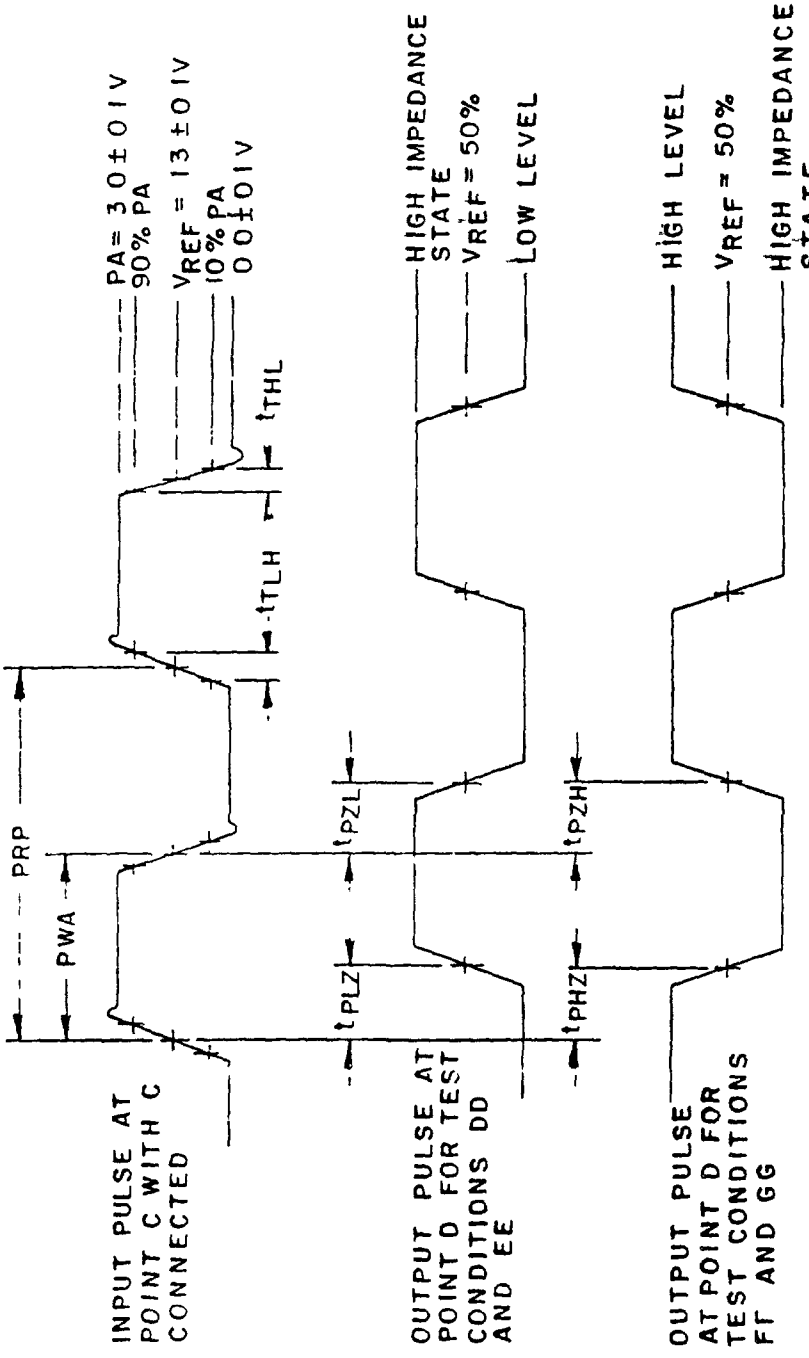
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NOTES
1. R1 = 51Ω ±5%, 1/2 W.
2. Load all Y outputs per figure 13, note 1.

*FIGURE 36. Enable recovery time (I4-I0 to Y, t_{PZL} and t_{PZH}) and disable recovery time (I4-I0 to Y, t_{PLZ} and t_{PHZ}) test circuit for tables XXY through XXY!!!

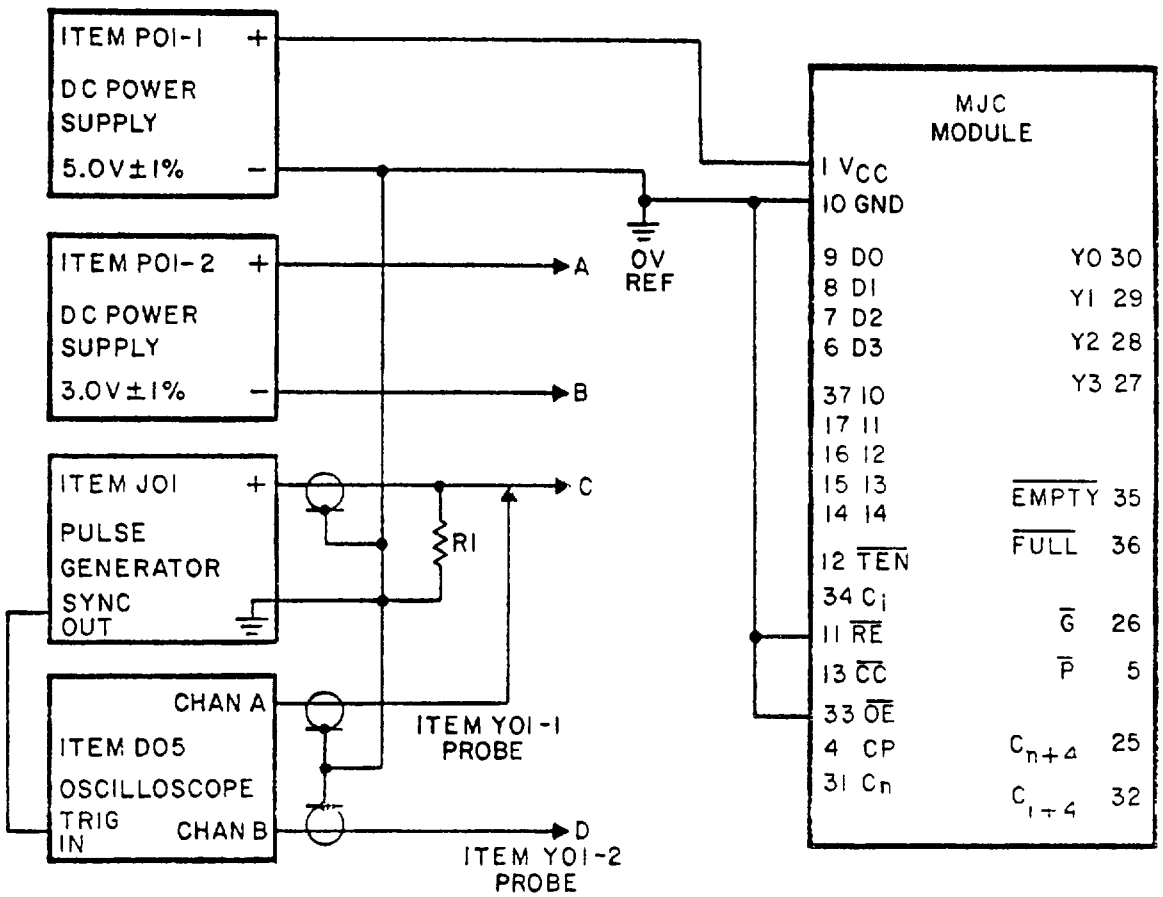
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NOTES:
1. Input pulse parameters:
   PWA = 500 ±50 ns
   PRP = 1,000 +100 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Before making specified measurements, verify that the output low level is ≤ 0.5 V and output high level is ≥ 2.4 V.

*FIGURE 37. Enable recovery time (I4-I0 to Y, t_{PZL} and t_{PZH}) and disable recovery time (I4-I0 to Y, t_{PLZ} and t_{PHZ}) test circuit waveforms for tables XXX through XXXIII.

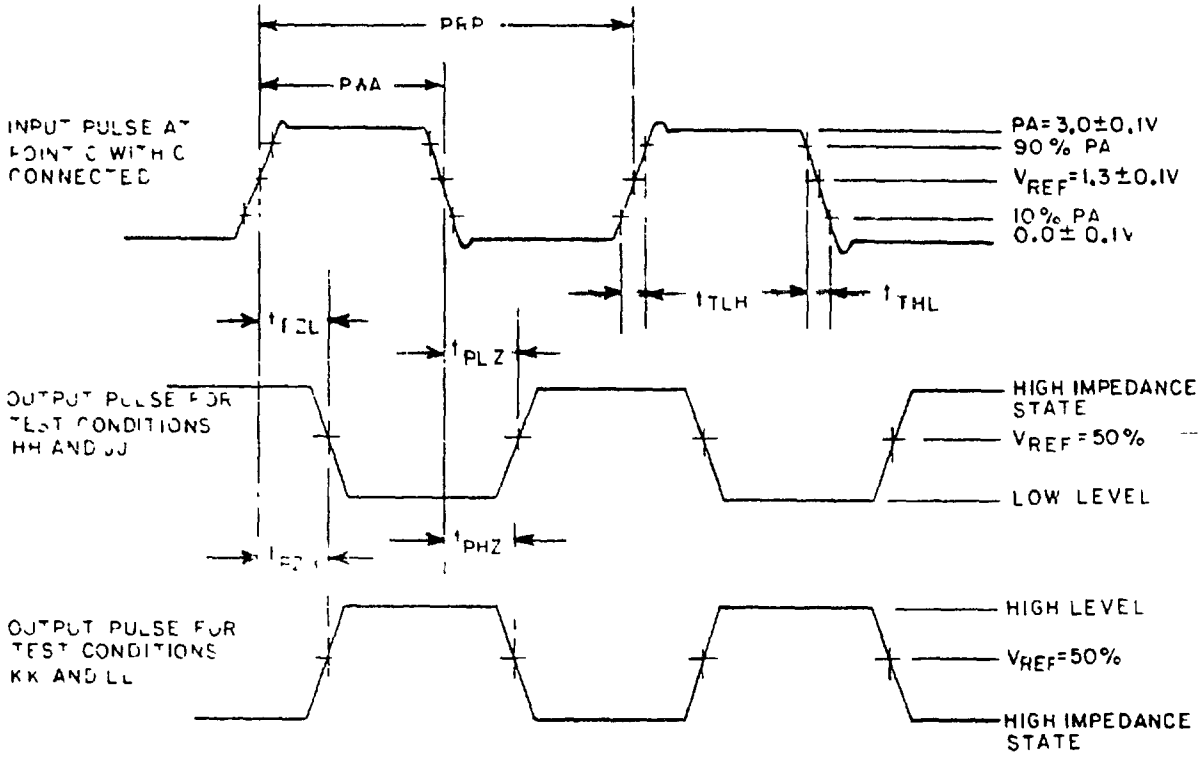
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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. Load outputs per figure 13, note 1.
3. Item J01 setup. Place in single pulse mode and clock module one time.

*FIGURE 38. Enable recovery time ($\overline{CC}$ to Y, t_{PZL} and t_{PZH}) and disable recovery time ($\overline{CC}$ to Y, t_{PLZ} and t_{PHZ}) test circuit for tables XXXIV through XXXVII.

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NOTES
1. Input pulse parameters
   PWA = 500 ±50 ns
   PPP = 1,000 ±100 ns
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Before making specified measurements, verify that the output low level is ≤ 0.5 V and output high level is ≥ 2.4 V.

FIGURE 39 Enable recovery time ($\overline{CC}$ to Y, t_{PZL} and t_{PZH}) and disable recovery time ($\overline{CC}$ to Y, t_{PLZ} and t_{PHZ}) test circuit waveforms for tables XXXIV through XXXVII.

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